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Innovation Management Systemicity

How Systemic Dimensions of Innovation Management Influence Innovation Capabilities

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Systemiskhet i innovationsledning

Hur systemiska dimensioner av innovationsledning påverkar innovationsförmåga

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SAMMANFATTNING

Trots att det finns mycket kunskap om innovationsledning har det funnits en tendens att studera olika aspekter inom området separat snarare än med ett systemperspektiv, vilket har resulterat i att viktiga samband mellan olika aspekter potentiellt har förbisetts. Systemperspektiv har använts i över 50 år inom organisationsforskning men har på senare år fått ökad relevans inom forskningen om innovationsledning. Ett tecken på denna ökande relevans är skapandet av en internationell vägledande standard för innovationsledningssystem, ISO 56002, som publicerades 2019. Forskning har visat att implementering av innovationsledningssystem genom att använda en standard eller inte, har underlättat för organisationer att övergå från ad hoc till mer integrerade innovationsledningsmetoder. Därför är det intressant att studera hur innovationsförmåga påverkas av att ta en systemansats till innovationsledning. Vidare kan hantering av radikal och inkrementell innovation kräva olika tillvägagångssätt, vilket skapar utmaningar, särskilt för stora företag eftersom de prioriterar projekt som oftast inkluderar inkrementell innovation. Därför är det relevant att studera hur inkrementell respektive radikal innovation påverkas av att hanteras inom samma system. I denna studie används begreppet systemiskhet (eng: systemicity) för att beskriva hur systemiskt ett system är genom tre systemdimensioner: omfattning (eng: comprehensiveness), sammankoppling (eng: coherence) och överrensstämmelse (eng: correspondence), vilka överlappar med dimensioner som vanligtvis används i litteraturen för att beskriva och definiera ett system.

Syftet med studien var att få en bättre förståelse för innovationsledning ur ett systemperspektiv genom att undersöka hur olika systemdimensioner påverkar och relaterar till innovationsförmåga, analysera dessa förmågor ur ett systemperspektiv, utforska hinder och möjligheter med en systemansats, samt undersöka förhållandet mellan dessa systemdimensioner och inkrementell respektive radikal innovationsförmåga.

Studien ingick i ett forskningsprojekt med det övergripande syftet att undersöka viktiga framtida utvecklingstrender och utvärdera den svenska metallindustrins innovationsförmåga för att använda som grund för att ta fram strategier för hur industrin kan stödja omställningen mot hållbarhet. En flerfallsstudie av två företag inom metallindustrin genomfördes med en abduktiv metod, inklusive en litteraturstudie, datainsamling, kodning och analys. Semistrukturerade intervjuer användes, där intervjufrågorna baserades på ISO 56002 och dess sju systemelement. Analysen gjordes i två steg genom att först analysera varje företag för sig och sedan jämföra de två fallen med varandra.

Resultaten visar på vikten av att ha förmågor relaterade till alla sju systemelementen för en organisations övergripande innovationsförmåga, vilket lyfter fram betydelsen av dimensionen omfattning. Sammankopplingarna mellan element framhäver betydelsen av systemdimensionen sammankoppling, och att etablera överensstämmelse mellan förmågor och mål förbättrar innovationsförmågan. Även om studien pekar på att det är fördelaktigt för en organisation att uppvisa omfattning, sammankoppling och överrensstämmelse i sitt arbete med innovationsledning, kan resultaten inte användas för att dra slutsatser om vilken nivå av systemiskhet som är lämplig. Därför behövs ytterligare forskning för att fastställa balansen mellan systemiskhet och flexibilitet för att effektivt stödja innovationsförmåga. I relation till att hantera inkrementell och radikal innovation tillsammans, visar studien på vikten av att ta systemiskhet i beaktning för att identifiera möjligheter och hinder vid hantering av båda typerna av innovation i samma system. Analysen av omfattning belyser behovet av olika hantering och olika innovationsförmågor i relation till varje innovationstyp. Analysen av sammankoppling betonar vikten av att studera beroenden mellan systemelement för att undvika att hinder som uppstår av kopplingarna blir förbisedda. Analysen av överrensstämmelse föreslår separata målsättningar för inkrementell och radikal innovation på grund av deras distinkta mål och resultat. Baserat på metoden och resultaten har ett tentativt ramverk tagits fram i syfte att användas för analys och utvärdering av systemiskhet i innovationsledning.

Nyckelord: Systemiskhet i innovationsledning, Innovationsledningssystem, Innovationsförmåga, Radikal innovation, Inkrementell innovation

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ABSTRACT

The field of innovation management has gained extensive knowledge; however, there has been a tendency to study its various aspects in isolation rather than with a systems perspective, resulting in a potential oversight of interconnections between important aspects. While systems perspectives have been employed in organization and management research for over half a century, systems perspectives have more recently gained relevance in innovation management research. A sign of the growing relevance is the creation of an international guidance standard for innovation management systems, ISO 56002, published in 2019. Research indicates that implementing innovation management systems, with or without using a standard, has facilitated the transition from ad hoc practices to more integrated ones. Thus, it is interesting to investigate how systems approaches influence innovation capabilities. Furthermore, managing radical and incremental innovation may require different approaches, which presents challenges, particularly for large companies that prioritize projects typically involving incremental innovation. Consequently, studying how incremental and radical innovation are influenced when managing both types within the same system becomes relevant. In this study, the concept of systemicity was used to describe how systemic a system is through three system dimensions comprehensiveness, coherence and correspondence, which overlap with those commonly used in literature to describe and define a system.

The purpose of the study was to gain a better understanding of innovation management from a systems perspective by investigating how different systemic dimensions influence and relate to innovation capabilities, analysing these capabilities from a systems perspective, exploring hindrances and opportunities with a systems approach, and investigating the relationship between these systemic dimensions and incremental and radical innovation capabilities respectively.

The study was part of a research project with the overarching aim to investigate important future development trends and evaluate the innovation capabilities of the Swedish metallic materials industry

to use as a basis to develop strategies for how the industry can support the transition towards sustainability. A multiple-case study of two companies within the metallic materials industry was conducted using an abductive approach, including a literature study, data collection, coding, and analysis. Semi-structured interviews were employed, where interview questions were based on the ISO 56002 standard and its seven system elements. The analysis involved within-case and cross-case analysis.

The findings from the study imply that having capabilities related to all seven system elements seem to be important for an organization's overall innovation capability, highlighting the significance of comprehensiveness. The interconnections between elements highlight the significance of coherence, and establishing correspondence between capabilities and goals enhances innovation capability. While exhibiting comprehensiveness, coherence, and correspondence is beneficial, the suitable level of systemicity remains uncertain. Further research is needed to determine the balance between systemicity and flexibility for effective support of innovation capabilities. Furthermore, in relation to managing incremental and radical innovation together, the study indicates the importance of considering systemicity to identify opportunities and hindrances when managing both types in the same system. The analysis of comprehensiveness highlights the need for distinct management approaches and capabilities for each innovation type. The analysis of coherence emphasizes the importance of considering interdependencies between system elements to avoid obstacles resulting from interdependencies being overlooked. The analysis of correspondence suggests separate goal setting for incremental and radical innovation due to their distinct objectives and outcomes. Based on the method and findings, a tentative framework was developed for analysis and evaluation of innovation management systemicity.

Keywords: Innovation Management Systemicity, Innovation Management Systems, Innovation Capability, Radical Innovation, Incremental Innovation

FOREWORD

During the spring of 2023 we, students at KTH Royal Institute of Technology, have conducted our master's thesis together with RISE Research Institutes of Sweden. We would like to thank everyone involved in providing help, assistance, cooperation and inspiration for our master's thesis. Thank you for the valuable assistance in putting all the pieces in place, making it come together, and guiding us towards achieving our goals. Furthermore, we would like to extend a special thank you to the following individuals for their contributions during the process.

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NOMENCLATURE

The abbreviations used throughout the study are presented here.

Abbreviations

IMS	Innovation Management System
IPO	Innovation Performance Objective
KPI	Key Performance Indicator
NPD	New Product Development
R&D	Research and Development

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1 INTRODUCTION

This chapter outlines the background of the study, its purpose and the delimitations of the thesis.

1.1 Background

Innovation management has amassed a significant body of knowledge, however, its impact is constrained by a lack of coherency (Tidd and Bessant, 2018). Seminal works have highlighted the significance of embracing a holistic perspective to comprehend complex problems and a system's performance as a whole (Van De Ven, 1986; Senge, Hamilton and Kania, 2015), recognizing that changes in one system element can impact the entire system in ways that are difficult to predict (Simon, 1947). System perspectives, or the practice of viewing systems as consisting of interconnected and interdependent elements and conceptualizing organizations as open systems that are influenced by their surrounding contexts, have been extensively employed in organizational and management research for over half a century (Katz and Kahn, 1966; Scott and Davis, 2007).

Research indicates that innovation output and performance are influenced by various factors such as organizational structure (e.g., O'Reilly and Tushman, 2004), organizational culture (e.g., Shore, Cleveland and Sanchez, 2018), and the innovation process (e.g., Cooper, 2008; Dorst, 2011). When studying innovation management, the focus has often been narrow, studying the aspects in isolation or with a reductionist scope, neglecting holistic perspectives which could result in suboptimizations (Magnusson, Kristiansen and Bessant, 2022). Companies generally struggle with resource allocation and prioritization in relation to innovation activities (Nilsson *et al.*, 2012) and could therefore benefit from a better understanding of how to prioritize to efficiently manage innovation, thus avoiding suboptimizations. This points to the need for further research and development in this field.

More recently, systems perspectives have become more common also within the field of innovation management and a sign of this increasing focus is the creation of an international standard for innovation management systems. In 2019, the International Organization for Standardization published a guidance standard for innovation management systems, ISO 56002, as part of the ISO 56000 series of standards for innovation management (The International Organization for Standardization, 2019). The implementation of innovation management systems, whether using a standard or not, has facilitated organizations in transitioning from ad hoc innovation management practices to more integrated ones (Hyland, Karlsson and Kristiansen, 2022), emphasizing the importance of evaluating innovation management systems to comprehend their status and potential improvements.

Innovation audits are used to analyse and evaluate innovation capabilities and performance of organizations. The various auditing approaches existing today utilize different dimensions to evaluate innovation performance (Björkdahl and Holmén, 2016), covering multiple elements considered in

proposed innovation management systems. However, existing auditing approaches fall short in sufficiently addressing the interrelationships of system elements and the overall innovation capabilities and performance. Consequently, they may not capture the connections between an innovation management system's performance and how systemic a system is.

While a systems approach to innovation management seems to be beneficial, researchers have pointed out different prerequisites for managing radical and incremental innovation (e.g., McDermott and O'Connor, 2002; Chang *et al.*, 2012; Tidd, 2022). Large corporations tend to invest in extensively planned and researched projects that promise high profits, often prioritizing their core business offerings, which typically involve incremental innovation (Bhidé, 1999). Eisenhardt et al. (2010) imply that efficiency tends to over-rule the flexibility needed for innovation in one and the same system and that approaches aimed to favour innovation thus need to accentuate flexibility rather than efficiency. If companies ignore to invest in ways to manage radical innovation activities, they will be ill-equipped when facing rapid changes in their environment (Kristiansen and Gertsen, 2015). Therefore, it is interesting to investigate whether using the same system for both types of innovation could have undesirable effects.

1.2 Purpose

The purpose of this study is to gain better understanding of innovation management from a systems perspective and to examine how various systemic dimensions impact a company's innovation capabilities. Furthermore, investigating different systemic dimensions in relation to both radical and incremental innovation capabilities is of interest as organizations, as outlined above, appear to face challenges in managing both types of innovation. By exploring this, the study aims to provide insights into possible hindrances and opportunities that might exist within the organization's innovation approach as well as how systemic innovation approaches can be analysed.

1.3 Delimitations

The study examines the systems perspective at the firm level, thus excluding an analysis of the entire value chain or ecosystem. Moreover, the study specifically focuses on companies operating within the same industry. Although the research centers on the companies' operations within Sweden, it is important to note that since these are international companies, some of the informants were located outside of Sweden.

2 EXPOSITION OF THEORY

To be able to explore the systems approach to innovation management and investigate the potential for managing both radical and incremental innovation within the same system, it was essential to draw upon prior research encompassing systems theory, innovation management and innovation capabilities (both incremental and radical). The theoretical base that forms the foundation for the study is visualized in Figure 2.1.

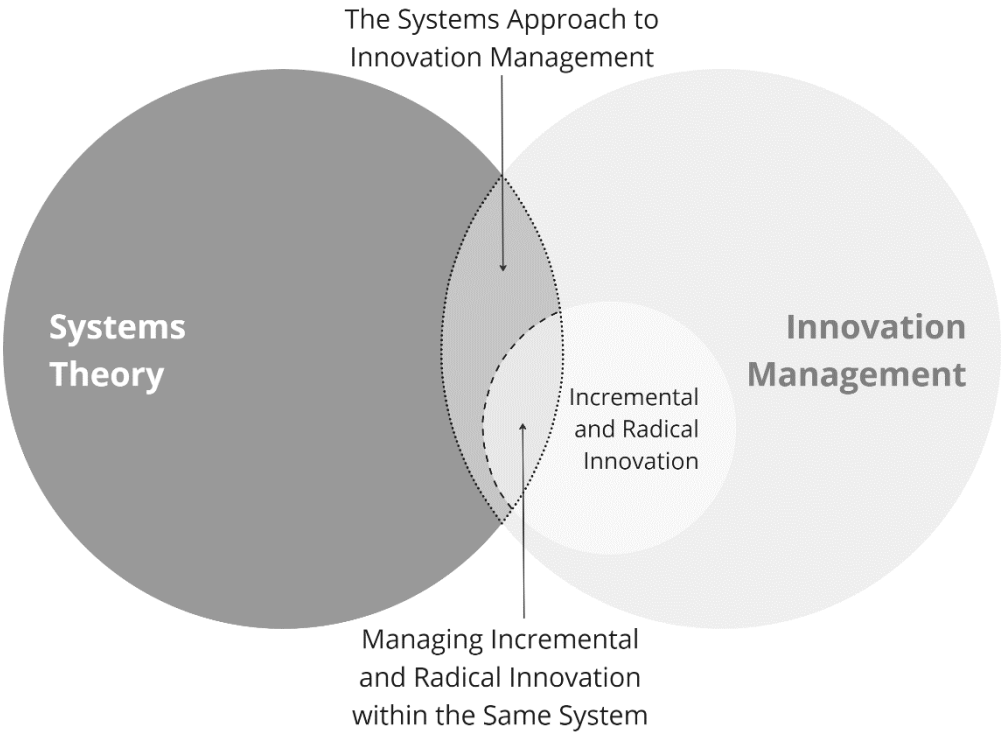


Figure 2.1. Illustration of theoretical base for the study.

2.1 Systems Theory

Systems theory provides a theoretical framework that describes systems as groups of interrelated and interdependent components, which are best studied together rather than separately (Wilkinson, 2011). Seminal works have underlined the significance of adopting a holistic perspective when addressing intricate problems and a system's performance as a whole (Van De Ven, 1986; Senge, Hamilton and Kania, 2015). These works emphasize the need to consider the interconnectedness and interdependencies among various elements within a system, recognizing that understanding the system as a whole is essential for comprehending its functioning and dynamics. This perspective acknowledges that changes in one element of a system can have effects on the entire system (Simon, 1947). Systems can vary in complexity and type, such as mechanical or social systems. The principles of systems theory have been applied in systems engineering (Walden *et al.*, 2015), but also in organizational and management theory, viewing organizations as systems composed of interactive subsystems (Wilkinson, 2011) as well as open systems that are influenced by their external contexts (Katz and Kahn, 1966; Scott

and Davis, 2007). Systems theorists acknowledge that systems can be hierarchically structured, which refers to that systems can be composed of smaller subsystems while also functioning as a subsystem within a larger system (Scott and Davis, 2007). Therefore, the system studied, or the unit of analysis, is defined by the delimitations set by the observer. Studying the interrelations between the subsystems of an organization as well as between the elements within each subsystem can help evaluate system performance and identify performance gaps (Karlsson and Magnusson, 2019).

2.1.1 Systemic Dimensions and Systemicity

Literature has described several aspects for analysing systems in general, including dimensions to analyse system performance of organizations. The concept systemicity, rooted in systems theory, has been used to describe systemic properties of systems to evaluate how systemic systems are. It has been employed in various fields such as project management risk (Williams, 2017), business process management (Segatto, Inês Dallavalle de Pádua and Pinheiro Martinelli, 2013) and biology (Baylis, 2000; Sgolastra *et al.*, 2020). Primarily, the concept has focused on exploring the interconnections and relationships among the elements within a system. Scott and Davis (2007) identified three dimensions covered in systems theory as fundamental when analysing systems performance. The first one, referred to as complexity or diversity, relates to the number of elements included and active simultaneously. The second dimension, uncertainty or unpredictability, addresses the degree of variability exhibited by the system elements, which can pose challenges in predicting system dynamics. The third dimension, interdependence, focuses on the interconnectedness among system elements. The authors distinguish between different levels of interdependence. Elements may be connected through a common overarching goal. Additionally, some elements may depend on others to fulfil their functions, while others may exhibit mutual dependence, where their performance relies on each other.

Although several aspects are considered in theory to understand system properties and dynamics, the structure and function of systems are commonly described using three dimensions, reflecting the aspects that define systems: (1) their components, (2) the connections between components, and (3) the common goals or purposes of the components (Segatto, Inês Dallavalle de Pádua and Pinheiro Martinelli, 2013). These dimensions align with the dimensions diversity, interdependency through a common overarching goal, and interdependency through function dependability described by Scott and Davis (2007). Furthermore, these dimensions are implicitly reflected in the well-established framework by Slack and Lewis (2017) for evaluating operations strategy, known as (1) comprehensiveness: the range of key decision areas, (2) coherence: the decisions made in each area do not have a counteracting effect on the direction of operations, and (3) correspondence: the strategies employed in each decision area should align with the actual priority of each performance goal. In the context of operations strategy, key decision areas refer to specific aspects or domains where key decisions need to be made, e.g., supply network or process technology. Performance goals refer to goals that an organization aims to achieve in its operations, e.g., flexibility or speed.

Systemicity appears to be an appropriate unifying concept to capture the various dimensions that determine how systemic a system is. However, as previously mentioned, the primary focus of systemicity has predominantly revolved around the interconnections among system elements. Given the theoretical emphasis on incorporating dimensions that encompass the entirety of elements within the system and their shared goals, it becomes evident that analysis and evaluation of systemicity should consider not only the interconnections, but also the comprehensive range of elements within the system and their collective objectives. These three aspects are covered by the three system dimensions comprehensiveness, coherence and correspondence outlined previously.

2.2 The Systems Approach to Innovation Management

Innovation management encompasses the process of managing new ideas from generation to implementation, including aspects such as strategy, process, and performance evaluation. Its objective is to create good conditions for organizations to innovate and respond to internal and external opportunities (Goffin and Mitchell, 2017). Several researchers have highlighted that in an intra-firm perspective, innovation management and operations management can be viewed as two parallel subsystems existing within the firm's management system. How these subsystems interact and are integrated depend on the organizational structure (Burgelman, 1983; March, 1991; Karlsson and Magnusson, 2019).

Researchers such as Midgley and Lindhult (2021) emphasize the significance of taking a systemic approach to innovation. They argue that systemic innovation goes beyond individual innovations and focuses on transforming the entire system. This approach requires systemic thinking, collaborative approaches, and an understanding of system dynamics to address complex challenges and drive sustainable change. Furthermore, the researchers argue for five different strands of thinking in systemic innovation literature. Firstly, systemic innovation is characterized by the convergence of complementary innovations within a larger product-oriented system, primarily driven by technology. Secondly, it encompasses regional policy systems that aim to support innovation within specific geographic areas. Thirdly, systemic innovation is seen as a powerful catalyst for achieving sustainability objectives. Fourthly, it emphasizes the importance of collaboration within innovation networks and ecosystems. Lastly, systemic innovation involves a shift in thinking, where the interconnectedness and interdependencies of various components within a system are considered. The purpose of this study primarily revolves around the fifth and last strand of systemic innovation, which Midgley and Lindhult (2021) emphasize as particularly important as it can support the other four strands in literature.

2.2.1 Innovation Management Systems

The literature on innovation audits partly embraces a system perspective by examining various components of innovation management, relating to the system dimension comprehensiveness. Tidd and Bessant (2013) highlight that innovation audits, assessing a firm's innovation management capability, provide insights into how effectively the firm manages innovation. In an article about innovation audits, Björkdahl and Holmén (2016) presents various audits existing in literature. These

audits reveal common elements to consider when analysing a firm's innovation capability, such as strategy and leadership, knowledge and learning, understanding the internal and external context, processes, culture, and planning. Similarly, Frishammar et al. (2019), Adams et al. (2006) and Nappi and Kelly (2021) discuss similar aspects, which indicates the importance of measuring several aspects of innovation management and the existence of a shared perspective in the auditing literature. However, one notable gap in this literature is the absence of a systems perspective as coherence and correspondence between elements are not considered. If viewing these aspects as elements of a system, they can be linked to the frameworks for innovation management systems (IMS) introduced over the years by various researchers from both academia and practice, as well as from national and international standards. Karlsson and Magnusson (2019) identified common elements across 19 selected frameworks that are crucial for organizations to achieve successful innovation. They summarize these in eight themes: context, direction, leadership, culture, processes, structures, support and resources, and evaluation.

2.2.2 Development of Standards

In practice, there have been a development of national and international standards for innovation management, reflecting the growing adoption of holistic approaches in this field. Standardized innovation management systems could accelerate the process from strategy development to effective implementation (Adams, Bessant and Phelps, 2006). The Spanish Association for Standardization and Certification (AENOR) developed the first innovation management system standard, UNE 16002:2006, which has been adapted and implemented by other countries, including Portugal, Mexico, Brazil, and Denmark. This was soon followed by other national standards, such as the BS 7000-1:2008 in the UK (Idris and Durmuşoğlu, 2021).

In 2019 the International Organization for Standardization (ISO) released parts of the ISO 56000 series of standards for innovation management, including ISO 56002, a guidance standard for innovation management systems (The International Organization for Standardization, 2019). The aim of ISO 56002 is to assist organizations involved in innovation management by providing guidance on determining their vision, strategy, policy, and objectives, as well as establishing the necessary support and processes to ensure that the intended outcomes are achieved. The standard serves to improve the quality and effectiveness of innovation work within organizations. It is designed to be applicable to all types of organizations, regardless of size, sector, or level of maturity in innovation management. The standard covers various aspects of innovation management through its seven elements: Context of the Organization, Leadership, Planning, Support, Operation, Performance Evaluation and Improvement.

While the implementation of standards is still limited, there is ongoing progress in this area, reflected in the increased focus on the systemic approaches and standards for innovation management within research, for example covering how standardization influence innovation capability and performance (e.g., Mir, Llach and Casadesus, 2022) and models to evaluate innovation management systems (e.g., de Carvalho Botelho Santos and Ludovico de Almeida, 2022). An example of the latter is a self-assessment model developed with the purpose of assisting established science and technology institutions in identifying crucial challenges and opportunities to enhance their innovation management capabilities

(de Carvalho Botelho Santos and Ludovico de Almeida, 2022). However, the self-assessment model primarily focuses on evaluating the current maturity level of an organization's innovation capabilities and does not include consideration of other dimensions of systemicity.

2.2.3 Opportunities and Hindrances with a Systems Approach to Innovation Management

Employing systems and systematic approaches towards innovation management through developing and implementing IMSs or applying innovation management standards have proven to facilitate organizations to move from ad hoc innovation practices to more conscious and integrated ones (Hyland, Karlsson and Kristiansen, 2022). In a multiple-case study presented by Hyland et al. (2022) both organizations that explicitly used the ISO 56002 standard and organizations that used other systems approaches were studied. In contexts where standards and certifications are regarded important, the use of the standard was shown to provide legitimacy. In other cases it supported a common understanding of the innovation activities carried out (Hyland, Karlsson and Kristiansen, 2022). Studies also imply that systems approaches and innovation management standards have supported in developing or improving innovation capabilities and contributed to improved business performance (Mir, Casadesús and Petnji, 2016; Hyland, Karlsson and Kristiansen, 2022; Mir, Llach and Casadesus, 2022). However, there are also studies that have reported on shortcomings of the standards and difficulties with adopting systemic innovation approaches. In the multiple-case study by Hyland et al. (2022) the findings from some cases suggest that the applicability to different types of organizations varies and furthermore, that challenges can arise in relation to deciding the scope of an IMS (Hyland, Karlsson and Kristiansen, 2022). These aspects have also been brought up by Tidd (2021), who question the applicability of the ISO standard in different organizational contexts (type, size and structure of organization, and industry sector) and for different types of innovation both regarding forms of innovation (e.g. product, service) and magnitude (incremental, radical). In two of the cases from the multiple-case study, organizations working with healthcare (Kihlander and Gravenius, 2022) and humanitarian aid (Bessant, 2022) were studied. These studies shed light on the unique demands and challenges faced by these particular contexts, emphasizing the need for innovative approaches distinct from those employed in industrial settings. However, the research also revealed the value of adopting a systemic approach, indicating that while adjustments may be necessary, a complete departure from common approaches might not be required. Some organizations examined in the study operate within a context characterized by extensive external collaborations (Bessant, 2022; Vasell and Nilsson, 2022). This resulted in a discussion on the potential benefits of implementing a shared IMS within the organizational ecosystem. In relation to deciding the scope of the IMS, some companies studied also exhibited difficulties implementing a universal innovation management system for the whole organization as there were large variations in the work conducted between different parts of the organization (Bensnes, 2022). Furthermore, one of the cases demonstrated the difficulties of a common approach to managing both incremental and radical innovation (Kristiansen and Witthaut, 2022). This study highlighted the distinction between the processes and tools required for managing radical and incremental innovation, which suggests that different systems may be needed for managing different types of innovation.

2.2.4 Managing Incremental and Radical Innovation within the Same System

Innovation capabilities refer to the collective traits and characteristics of an organization that enable and foster the development and implementation of innovation. These capabilities encompass a wide range of factors, such as the organization's skills, resources, processes, culture, and strategies, which contribute to its ability to generate and support innovative ideas, products, or services. Innovation capabilities vary depending on the type of innovation a firm is engaged in, whether it is incremental or radical. Incremental innovation tends to focus on efficient, exploitative and aligning activities, while radical innovation mainly emphasizes exploration, flexibility-enhancing, and adaptive activities (Chang *et al.*, 2012). Lennerts et al. (2020) expresses the ineffectiveness of combining exploration with exploitation for radical innovation, compared to incremental innovation that can benefit from a combination of the two.

The differences in innovation capabilities required to manage incremental and radical innovation pose challenges for companies in jointly managing both types, however, both types are valuable as they contribute to value creation (O'Reilly and Tushman, 2004). Nilsson et al. (2012) discuss these challenges and summarize them as four dichotomies: time, uncertainty, flexibility and control. The authors argue that radical innovation involves adopting longer-term perspectives and navigating discontinuity, dealing with higher degrees of uncertainty, establishing adaptable structures, processes, and strategies to facilitate change, and fostering a greater degree of freedom compared to incremental innovation.

Some of the dichotomies presented by Nilsson et al. (2012) primarily address the means to achieve radical innovation rather than objectives to aim for. To assess if a company's actions align with the goals required for implementing radical innovation, it can be advantageous to express the dichotomies: time, uncertainty, flexibility and control, as factors in a way that allows for performance evaluation of management of radical innovation. Utilizing the insights about these dichotomies, three performance objectives necessary for radical innovation can be formulated, firstly time, covering whether long-term perspectives are considered, secondly risk-taking, encompassing if there is a willingness to take risks, and thirdly flexibility, examining whether flexibility is enabled. In addition to these three performance objectives, prior research highlights the significance of considering novelty and cost, as established firms tend to invest in and prioritize activities related to incremental innovation (Bhidé, 1999).

While a systemic and systematic approach may be suitable for incremental innovation, where current knowledge is largely utilized (Tidd and Bessant, 2013), the same approach may not be effective for radical innovation, where risks and uncertainties are higher, and variations are more fundamental (McDermott and O'Connor, 2002; Reinertsen and Shaeffer, 2005). However, managing uncertainties and risks could be supported by structures such as those brought up in innovation management systems literature (Karlsson and Magnusson, 2019). The dichotomies or tensions in managing incremental and radical innovation point to the relevance of investigating whether using the same system for both types of innovation could have undesired effects. The five innovation performance objectives time, risk-

taking, flexibility, novelty and cost, could support analysis of potential conflicts in jointly managing the two types.

2.3 Research Questions

Based on the purpose of gaining better understanding of innovation management systemicity and its impact on a company's innovation capabilities, the following two research questions have been developed:

> RQ 1: How does innovation management systemicity influence an organization's innovation capabilities, where systemicity is defined by the three dimensions (1) comprehensiveness, (2) coherence, and (3) correspondence?
>
> RQ 2: What does systemicity indicate regarding hindrances and opportunities for managing incremental and radical innovation within the same system?

3 METHOD

The method chapter comprises several subchapters outlining the research approach, research setting, literature study, data collection, data analysis, and research quality.

3.1 Research Approach

How innovation management systemicity influences an organization's innovation capabilities, was explored in a multiple-case study of two established companies within the metallic materials industry. A case study approach was applied as it is suitable in contexts where the subject matter has not been studied much (Eisenhardt, 1989) and for exploring questions of how (Voss, Tsikriktsis and Frohlich, 2002). The study was conducted with an abductive approach in line with Dubois and Gadde (2002), continuously cycling between theory and empiricism. Furthermore, data collection, coding and analysis were conducted iteratively, which enabled consulting theory as new insights emerged from empirical findings and iterating back when new theoretical insights were discovered. Dubois and Gadde (2002) argue that an abductive approach is appropriate when exploring new aspects or subjects, developing, rather than confirming theory from existing knowledge or generating new theory.

Data was collected through semi-structured interviews to balance between the predefined structure of the interview guide and emergent concepts. To develop a richer understanding of contextual aspects and enable concepts outside the theoretical framework to emerge, coding and analysis of data was done initially taking a more data-driven, inductive approach, inspired by Gioia, Corley and Hamilton (2013). However, the process also included more theory-driven and abductive elements as empirical data was compared to theory in later stages. The cases were analysed using within-case, followed by cross-case, analysis in line with Eisenhardt (1989), to allow for identification of within-case patterns and initial development of theoretical concepts and thereafter comparing the cases and seeking cross-case patterns. In the succeeding sections of the chapter, the context of which the study was carried out is covered, followed by more detailed descriptions of how the literature study, data collection and data analysis were conducted.

3.2 Research Setting

This study was part of a research project in which the commissioner of the thesis, RISE (Research Institutes of Sweden), was one of the project partners. From the commissioner's perspective, the purpose of including the thesis in the research project was to contribute to the advancement of knowledge of the systems approach in the field of innovation management. The research project was part of Metallic Materials, one of the strategic innovation programs funded by Vinnova, Formas and the Swedish Energy Agency. The overarching aim of the research project was to investigate important future development trends and evaluate the innovation capabilities of the industry to use as a basis to develop strategies for how the Swedish metallic materials industry can support the transition towards sustainability. Companies are required to reduce their environmental impact to meet climate targets and

the metallic industry is facing significant transformations as it is a large contributor to greenhouse gas emissions. For example, the Swedish metal industry has during the past decade contributed to approximately 40 percent of the total national industrial greenhouse gas emissions (The Swedish Environmental Protection Agency, n.d.). The industry is mature and has a long history of innovation, traditionally developed during long cycles. Considering the significant changes and challenges within the industry needed to be solved during a short period of time, the context of the metallic industry seemed appropriate for selecting cases to study the influence of systemicity on innovation capabilities and managing incremental and radical innovation within the same system.

3.3 Literature Study

An initial literature study was conducted to develop a basis to support the data collection and analysis. Areas related to systems theory in an organizational studies context, innovation management systems and management of incremental and radical innovation were researched. In line with the abductive research approach, literature was continuously reviewed and consulted during the study (Dubois and Gadde, 2002). Additional literature was included based on suggestions provided by supervisors as well as searching the reference list of already included literature. Scopus and Web of Science were used as primary search engines in the literature search. Key search words are summarized in Table 3.1.

Table 3.1. Key search words in the literature study

Research Area	**Search Words**
Systems Theory	systemicity, systems analysis, systems dimensions, systems engineering, systems thinking, systems management
Innovation Management Systems	innovation management system, innovation management system standard, innovation standard, ISO 56000, ISO 56002
Innovation Management	innovation management, innovation audits
Radical and Incremental Innovation Capabilities	innovation capabilities, dynamic capabilities, resource-based view, incremental innovation, radical innovation, transformational innovation, breakthrough innovation, exploitative innovation, explorative innovation, ambidextrous innovation

3.4 Data Collection

Data was collected through semi-structured interviews to balance between the predefined structure of the interview guide and emergent concepts. Furthermore, it allowed for follow-up questions to probe areas where informants possessed greater expertise, enabling a more comprehensive exploration of the research topic. The following sections provide an overview of the selection of companies and informants, the design of the interview guide, and the data gathering process.

3.4.1 Selection of Companies and Informants

The informants were employees from two companies within the Swedish metallic materials industry. The companies were selected in close dialogue with the project management team of the research project within Metallic Materials to provide different perspectives on how innovation is managed to meet the challenges and opportunities that the industry is facing. The companies will be referred to as Company A and Company B. Company A is a leading global provider within their niche within the metal industry, specializing in the production and supply of high-quality material for various industries with a focus on the automotive industry. Company B is an established company within the metal industry that specializes in providing innovative solutions and advanced technologies for different industries like the automotive, battery and steel industries. The company are world leading within their area. Both companies act on the global market.

The informants were purposively selected in collaboration with company representatives, in line with the approach described by Voss et al. (2002). The company representatives were instructed to select informants based on specific preferences and requirements. The aim was to gather insights about the different aspects of innovation management and therefore the representatives were asked to select informants to cover roles with responsibility over project management, budget and resource allocation, strategic oversight of development and innovation, process management and innovation project oversight, responsibility for development work, and decision-making authority for project implementation. In addition to the top-down perspective, informants to cover the bottom-up perspective were also requested. The representatives' understanding of the organization allowed them to identify individuals who possessed comprehensive knowledge or specific insights into different aspects of the company's innovation management activities. Within Company A, multiple informants were chosen within a specific product area to gain in-depth insights. In addition, key roles from other product areas were included to ensure a comprehensive perspective across the entire organization.

3.4.2 Design of Interview Guide

The interview guide (see Appendix A) involved semi-structured questions, which provided flexibility for the informants to delve into areas they considered important and relevant. The interview questions were designed based on the ISO 56002 standard and its seven system elements: Context of the Organization, Leadership, Planning, Support, Operation, Performance Evaluation, and Improvements. The elements aim to capture a holistic understanding of the organization's innovation capabilities and practices. Context of the Organization refers to understanding the internal and external factors that

influence the organization's innovation efforts. Leadership focuses on the role of leaders in driving and supporting innovation within the organization. Planning involves developing a structured approach to innovation. Support encompasses the provision of necessary resources, capabilities, and infrastructure to enable innovation. Operation focuses on the execution and management of innovation initiatives. Performance Evaluation involves assessing the effectiveness and impact of innovation activities. Improvements emphasizes the continuous learning and refinement of innovation practices. The spread of the interview questions over the system elements is illustrated in Figure 3.1 below.

To test the interview guide, assess the clarity of the questions, and identify any potential issues or improvements, a pilot interview was conducted with an informant working with innovation as a Development Manager in an industrial company, separate from Company A or B. The interview guide was also revised in consultation with senior researchers active within the research area.

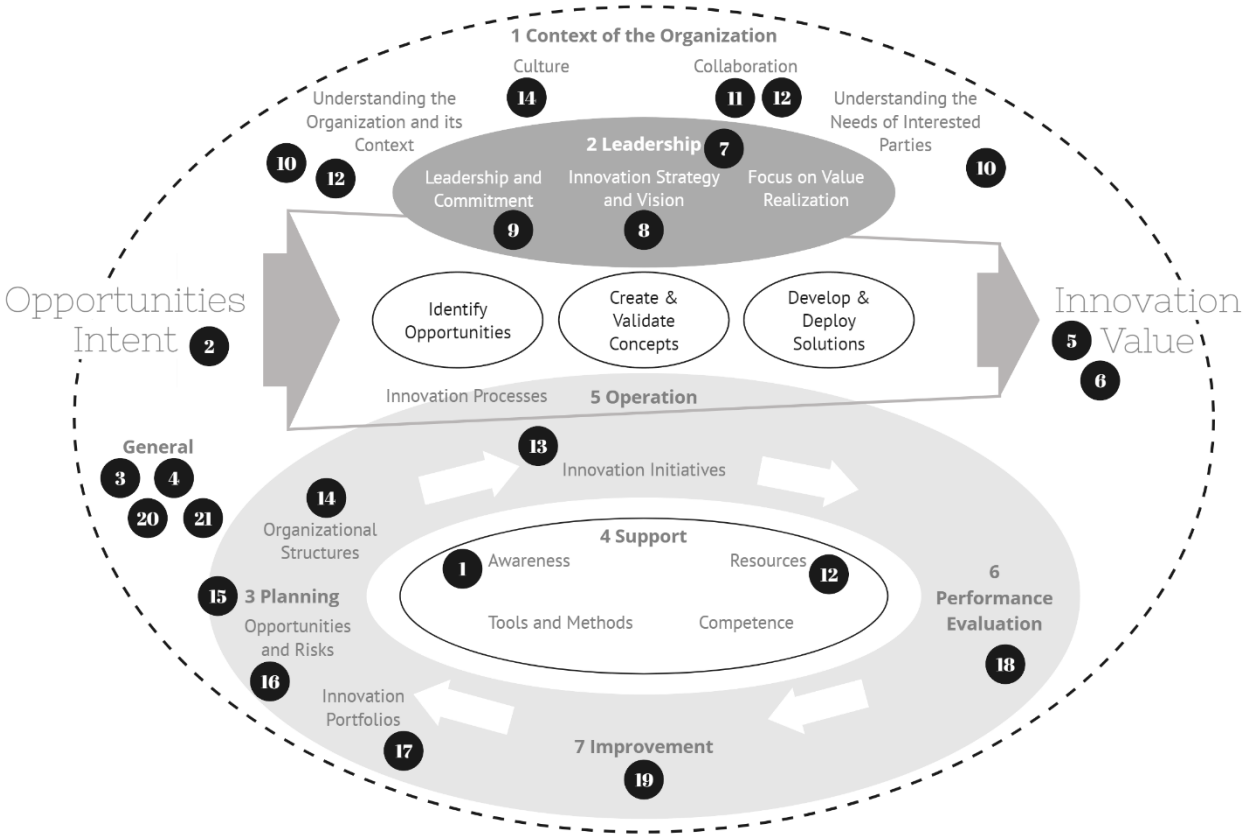


Figure 3.1. Coverage of interview questions in relation to ISO 56002. Figure based on an adoption of ISO 56002:2019.

3.4.3 Data Gathering

A total of 18 semi-structured interviews were conducted with employees from the two companies, 12 in Company A and 6 in Company B. The imbalance in number of interviews between the two companies resulted from limitations in availability, but informants were selected to cover equivalent roles within both companies. Six of the interviews at Company A were conducted in collaboration with a senior researcher, and four interviews at Company B were conducted in collaboration with a senior lecturer, both involved in the research project within Metallic Materials. The interviews lasted for 45-60 minutes and were recorded with the consent of the informants. The recordings were utilized for transcription purposes. Seven interviews were conducted with Company A in-person at their main office, while the five remaining interviews with Company A and all the interviews with Company B were conducted online via Microsoft Teams. The interviews are presented in detail in Table 3.2. Company A is organized in product or technical areas referred to as X, Y and Z.

Table 3.2. Information regarding informants and conducted interviews.

Company	**Role**	**Type**	**Length [min]**	**Date**
A	President in X	In-person	54	03-04-2023
A	Vice President in X	In-person	60	03-04-2023
A	Vice President in X	In-person	57	04-04-2023
A	Director in X	Digital	55	28-03-2023
A	Project Manager X	Digital	48	31-03-2023
A	Manager in X	In-person	48	03-04-2023
A	Engineer in X	In-person	58	03-04-2023
A	Manager in Y	In-person	45	03-04-2023
A	Director in Y	Digital	48	17-04-2023
A	Vice President in Z	In-person	55	03-04-2023
A	Managerial in Operations	Digital	58	31-03-2023
A	Managerial at Subsidiary	Digital	45	04-04-2023
B	Managerial R&D	Digital	51	06-04-2023
B	Managerial HR	Digital	60	28-03-2023

Company	Role	Type	Length [min]	Date
B	Manager	Digital	56	27-03-2023
B	Manager	Digital	50	18-04-2023
B	Product Manager	Digital	55	28-04-2023
B	Engineer	Digital	55	28-04-2023

3.5 Data Analysis

Coding of the data was done with an inductive approach inspired by Gioia, Corley and Hamilton (2013) and the analysis followed the approach of within-case and cross-case analysis in line with Eisenhardt (1989), starting with separate analyses for the two companies and followed by a joint analysis of the two.

3.5.1 Coding Procedure

The coding procedure followed several steps to organize and analyse the data and was done for each company separately. Coding was done using the qualitative analysis software ATLAS.ti. The first step of coding aimed to capture the essential aspects of the participants' responses without attempting to extract categories in line with Gioia, Corley and Hamilton (2013). Relevant information related to ways of working, obstacles, opportunities, weaknesses, strengths, and perspectives of systemicity in relation to innovation in general, or specifically related to incremental or radical innovation, were highlighted. Secondly, the highlighted information was summarized in first-order codes, which covered the responses and centred around the informants' answers. Additionally, if the responses were related to one of the standard elements, codes representing the system elements Context of the Organization, Leadership, Planning, Support, Operation, Performance Evaluation, and Improvement, were added to facilitate sorting and organization. This resulted in 450 first-order codes for Company A and 257 for Company B. Thirdly, the first-order codes within each overarching theme were reviewed to identify similarities and differences among the categories and were clustered into first-order concepts. This process, similar to axial coding (Gioia, Corley and Hamilton, 2013), aimed to reduce the number of categories to a more manageable level. The first-order codes were reduced to 30 first-order concepts for Company A and 18 for Company B.

3.5.2 Analysis

The analysis was conducted by first analysing the findings for each case and then comparing the two companies in a cross-case analysis. For the within-case analysis, the first-order concepts were first mapped and connected to the system elements in the ISO 56002 guiding standard. Thereafter, the findings were analysed by studying the concepts looking at the system dimensions comprehensiveness, coherence and correspondence. These dimensions were chosen based on their promising usefulness for

assessing the systemicity of innovation management approaches, as they have proven useful for evaluating operation strategy through the framework Operation Strategy Matrix developed by Slack and Lewis (2017).

The *comprehensiveness* dimension was used to analyze the extent to which the innovation capabilities of each company cover the seven elements related to the ISO 56002 standard, Context of the Organization, Leadership, Planning, Support, Operation, Performance Evaluation and Improvement, by considering both the spread over the elements and the maturity level of the capabilities. In some cases, assessing the maturity level of elements involved a two-step process: first assessing the maturity level of the areas within each element and then calculating the mean value of the included areas to evaluate the maturity level of the element as a whole. This approach was employed when there were significant variations in maturity levels among the areas within an element. For elements where the findings indicated a clear maturity level, this two-step process was not applied. Innovation capabilities in relation to the elements were assigned maturity levels from a three-point scale, ranging from one to three, according to the levels and corresponding description outlined in Table 3.3 below.

Table 3.3. Maturity levels, adapted from Jochem et al. (2011) and de Carvalho Botelho Santos and Ludovico de Almeida (2022)

Maturity Level	**Description**
Level 1: Nothing, informal, or ad hoc	The capability is either non-existent or established in an informal or ad hoc manner. The capability is managed in a deficient manner or lacks a clear definition.
Level 2: Managed at basic level	The capability is established at a basic level, meaning it is partially, but not fully, realized.
Level 3: Defined and managed	The capability is defined and established with a formal character, as well as actively managed.

The *coherence* dimension of the system was used to analyze the relationships between elements identified in the findings. Analysis of interconnections were based on concepts that explicitly demonstrated connections, and other potential connections within the data were not considered. The type of interconnections was also analyzed.

The *correspondence* dimension was used to analyze the alignment between the companies' actions and their goals. To conduct the analysis, innovation performance objectives related to managing both incremental and radical innovation were used. This approach was chosen considering the aim of the research project within Metallic Materials and the shared aim of both companies to be leading companies in the industry's transition towards sustainability. The system dimension correspondence is

particularly relevant for analysis as the transition necessitates radical innovation, while established companies often prioritize incremental innovation (Bhidé, 1999). The five innovation performance objectives (IPOs) employed for analysis include time, risk-taking, flexibility, novelty, and cost. In relation to the IPO time, correspondence was analyzed considering whether the companies included long term perspectives, often required to enable radical innovation (Nilsson *et al.*, 2012). The analysis of correspondence to the IPO risk-taking considered the companies' willingness to take risks and embrace higher degree of uncertainty, which is needed to facilitate radical innovation (Nilsson *et al.*, 2012). Correspondence to the IPO flexibility was analyzed considering whether the companies allowed for flexibility in their management of innovation, necessary for radical innovation (Nilsson *et al.*, 2012). The analysis of correspondence in relation to the IPO novelty focused on the level of novelty in the solutions developed by the companies. It sought to study the extent to which the innovations introduced by the two companies exhibited novelty in their offerings. Lastly, in relation to the IPO cost, correspondence was analyzed considering the resources allocated and invested into radical innovation by the companies. The last two IPOs were analyzed because established firms typically prioritize and invest in incremental innovation (Bhidé, 1999).

After analysis of each case separately, the cases were compared in a cross-case analysis considering the three system dimensions comprehensiveness, coherence and correspondence, with the purpose to study similarities and differences between cases, as well as to search for patterns (Eisenhardt, 1989).

3.6 Research Quality

Voss et al. (2002) highlights that it is especially important to pay attention to reliability and validity when conducting case study research. Reliability and validity, in turn, have multiple dimensions for evaluating research quality. According to Yin (2009), some of these dimensions are internal validity, external validity and reliability. The author further describes these dimensions as follows. Internal validity examines whether the steps of a study accurately address the research questions without bias. External validity concerns the generalizability of study results. Reliability relates to the consistency of results when a study is replicated. The case study tactics used to ensure validity and reliability for this study are summarized in Table 3.4.

Table 3.4. Case study tactics for ensuring research quality. Table adapted from Yin (2009).

Quality dimension	**Case study tactics**	**Phase of research in which tactic occurs**
Internal validity	• Looking for similarities and matches between the observed patterns and the expected patterns based on established theories	Data analysis

External validity	• To ensure comprehensive coverage, the informants were asked to provide recommendations on relevant informants • Linking results to literature	Data collection Data analysis
Reliability	• The diversity of informants provides multiple perspectives (including both bottom-up and top-down viewpoints) • Methodology of data collection and analysis are described in detail	Data collection Data collection

According to Eisenhardt (1989), having multiple researchers present during interviews increases the likelihood of discovering unexpected findings and enhances confidence in the data collection process. In this study, both authors were present during all interviews, and in more than half of them (10 out of 18), an additional researcher was involved. This approach enabled effective management of interview questions while facilitating comprehensive notetaking and observations as well as allowing for follow-up questions by all researchers. By adopting a collaborative approach, the study aimed to capture a more comprehensive understanding of the data, ensuring that valuable insights were not overlooked.

To ensure intercoder reliability, the two authors adopted a collaborative approach throughout the coding process. Initially, a test coding was conducted independently by each author to assess the consistency of their coding approaches. The authors then came together, reviewing and cross-checking their individual coding to identify any discrepancies and align their interpretations. Regular discussions were held to address uncertainties and promote a shared understanding of the codes, following the approach outlined by Gioia et al. (2013). This collaborative approach resulted in a high level of agreement and reliability among the two coders.

4 RESULTS

In presenting the findings, the results are organized into sections that align with the standard elements: (1) Context of the Organization, (2) Leadership, (3) Planning, (4) Support, (5) Operation, (6) Performance Evaluation, and (7) Improvements. The description in each section is based on the informants' answers during the interviews through the first-order concepts generated from the coding procedure. The concepts are specifically associated with one or two (or occasionally three) elements and therefore, findings in relation to the concepts are sometimes repeated to highlight interconnectedness to several elements as well as specific aspects related to the specific element under consideration. Where certain findings did not align with the standard elements, these are explained in chapters 4.1.8 and 4.2.8 for each company respectively. The concepts and their relation to the standard are presented and illustrated in the summarizing chapters 4.1.9 and 4.2.9.

4.1 Company A

Company A is a leading global provider within their niche within the metal industry. Specializing in the production and supply of high-quality materials for various industries with a focus on the automotive industry. In recent years, the company has undergone several changes, including shifts in management and a reevaluation of its approach to leading innovation activities. Notably, Company A is on a journey towards an outside-in approach, prioritizing customer needs in their innovation processes, compared to previously when the approach was more centred around an inside-out perspective, where the focus was on creating a desire for their products, without necessarily aligning with customer needs. This shift reflects their proactive response to the significant global movement towards sustainability. Recognizing the importance of environmental responsibility and sustainable practices, Company A has embarked on a transformative journey. By implementing a series of changes, they have aligned their operations with the principles of sustainability, demonstrating their commitment to being at the forefront of this movement. Amidst these changes, Company A has gained a renewed focus on driving innovation and delivering solutions that meet the evolving needs of their customers. The company operates its innovation activities (as well as line activities) under a matrix organizational structure, where three product areas are integrated with three sales units, each distributed across different continents. Additionally, an operations unit and supporting functions are integrated with all parts of the organization. The company does not have a centralized unit for research and development (R&D). Instead, the responsibility for R&D is distributed among each product area, as well as the operations unit.

4.1.1 Context of the Organization

Although there are several drivers of innovation within the company, one of the most prominent ones are customer needs, which they work to match with their internal competencies. They emphasize fostering close customer relationships to gain a competitive advantage and actively engage with customers. As an example, one purpose of creating a subsidiary focused on providing engineering

services was to enable improved insights into customer needs. Recognizing the importance of understanding end users' needs, they aim to deliver maximum value and minimize the risk of introducing solutions that do not meet the needs or expectations of customers. This is done through applying a total value approach, where they focus on higher product quality and improved performance, resulting in lower costs for customers. However, informants highlighted that strong customer focus may limit innovation and hinder exploring new business opportunities beyond their current value chains, potentially overlooking future customers and prospects.

Sustainability is another important innovation driver for the company, driven by both internal and external needs. There is a long-term focus on sustainability, driven by the belief that it will bring value in the future. While the sustainability shift poses challenges and requires additional resources, it was also expressed that it presents opportunities for innovation. Moreover, it was emphasized that a focus on sustainability is necessary for the company's long-term survival.

The company has established external collaborations and methods to identify external trends and drivers. The company participates in external financed projects, conferences, engages with customers, experts and utilizes collaborations and consultants to gather information from relevant areas outside their core business. To stay prepared for market changes, they proactively engage with literature, magazines, and participate in consortiums. While they are successful in collaborating with universities to leverage research findings and establish recruitment, informants mentioned that there has been a decrease in engagement into these collaborations after the management shift. Furthermore, there is currently no formal process to effectively utilize insights gained from external monitoring and collaborations. Sales are responsible for capturing trends and drivers, but there is a lack of structure in utilizing this information internally.

In general, the company faces challenges with internal collaborations and knowledge sharing. The company lacks a structured approach to knowledge sharing, mostly relying on informal contexts and individuals' efforts. Informants emphasized the importance of collaboration and knowledge sharing between different functions in the current matrix organizational structure as cross-functional collaboration is perceived to help address risks and uncertainties effectively. However, the company faces challenges in sharing knowledge across groups and product areas. Although teams are organized based on competence, there is a lack of interdisciplinary collaborations, hindering alignment towards shared goals.

The company's organizational culture currently does not embrace failure and there is an absence of necessary structures and support for learning from failures. However, the company acknowledge the importance of embracing failure as a mean of learning and value knowledge gained from such experiences. To foster a culture that accepts failure and encourages the expression of opinions, they understand that it is crucial to allocate sufficient time and resources to leverage the insights and ideas derived from failures.

The company is also currently facing challenges in relation to allocating resources. The difficulties with managing resources include a lack of support to plan resources as well as an overview of competencies. There is a recognized need for implementing a tool or support system to manage resources effectively, specifically to identify gaps and address existing difficulties. To maximize resource utilization, they emphasize the importance of aligning individuals with their respective strengths and placing them in appropriate positions.

4.1.2 Leadership

The company do not have a separate strategy for innovation, but the overall strategy guide innovation focus. This strategy is formed and communicated top-down, broken down to each product area or unit, where employees can contribute to and impact the strategy, which indicate that the strategy is not rigid and can be adjusted if considered necessary. The extent to which it is broken down varies within the company. Notably, one product area has implemented individual meetings and quarterly unit meetings to foster a shared understanding of the strategy and vision. However, this practice is not uniformly implemented across all product areas.

In line with having strategic flexibility, interviewees emphasize the importance of staying informed and remaining flexible to adapt the strategic direction. This allows for the inclusion of new insights and helps mitigate risks and uncertainties for ongoing success. The company's follow-up processes involve reflecting on external changes and recognizing the significance of continuous monitoring of the external environment, which can influence business plans. As a result, new insights from external monitoring and internal discussions can lead to the revision of business plans.

Top management is actively involved in setting overarching goals, establishing structures, and providing direction, although they may not directly participate in innovation activities. Product area management, on the other hand, plays a crucial role in continually following up on the strategy and remaining actively engaged in daily work. Interviewees believe that management bears the responsibility of offering direction and support for innovation while allowing employees the freedom to determine how it should be carried out. Management's role is hence to define the intent and focus of innovation, empowering employees to find the most effective approaches. While the company has clear and well-established direction, improvements are needed in providing support for innovation, such as time allocation and enabling collaboration, as reflected in the previous section.

It is recognized that bringing continuous value and ensuring survival require considering both short-term and long-term perspectives. Management is responsible to consider this balance when choosing the strategic direction. The shift from the historically long innovation cycles within the industry to the need of shorter innovation cycles connected to higher degrees of uncertainties has made the company chose to apply strategy looking at two perspectives, short-term and long-term. Mid-term perspectives are perceived to be too uncertain, why the choice of having flexibility in strategy has been made. Balancing these two perspectives are also reflected in the organizational matrix structure, where product areas focus on future perspectives, while the sales areas focus more on current aspects. Furthermore, the

importance of long-term perspectives is reflected in the steps the company takes to foster it, such as recruiting employees who align with this outlook.

4.1.3 Planning

As described in the previous section, the importance of balance between short-term and long-term perspectives is recognized within the company. From a planning perspective, this is related to the organizational structure. One of the company's rationales of organizing according to a matrix structure is that it enables capturing both short-term and long-term aspects. Furthermore, planning relates to formulating objectives to achieve high-level goals and targets in the strategy. This is done, as mentioned in the previous section, by breaking down strategy to a product area, group, and in some cases individual level.

Related to organizational structures is also how the company manage radical innovation when ideas have been matured, evaluated and decided to be commercialized or implemented. From what was captured in the interviews, the company mostly conduct incremental innovation, focused on improvements or smaller alterations of current solutions. When radical innovation, including exploring more novel solutions, is conducted, it is first studied within the existent organization. However, when the decision is made to pursue this type of innovation, it often necessitates a significant allocation of resources and typically prompts the establishment of new organizational structures or product areas to manage them.

The interviewees emphasized that both efficiency and innovation as well as to strike a balance between the two are crucial for continuous relevance and success. While there is an acknowledged need to allocate time and create room for innovation to foster creativity and exploration, it is perceived that the organization currently allocates limited time for these activities and that there is an emphasis on efficiency.

Top management receives continuous updates on projects and high-level activities. The product area which was studied more in depth has recently started to consider the totality of projects and activities, adopting a portfolio approach. They have now implemented systematic and continuous follow-up on projects and activities in which people from different functions and levels participate. The informants expressed that by establishing a shared foundation for tracking, planning, and decision-making in projects and other activities, a notable improvement in maintaining consistency in progress of these activities has been achieved.

One important aspect of planning in relation to risks and uncertainties that was highlighted in the interviews was how risks and uncertainties associated with external conditions such as raw material prices and availability are handled. These are perceived to be effectively addressed through proactive measures and flexibility. Proactive measures involve establishing backup options, such as alternative suppliers, to ensure a reliable supply chain. Additionally, actively monitoring market conditions and maintaining flexibility in finding solutions can provide a foundation for the company from which to navigate and respond effectively to potential risks and uncertainties that arise from external factors.

4.1.4 Support

There is currently no widespread understanding or use of common language in relation to innovation within the organization. While there is some consensus that innovation involves research, pre-studies, developing new products, and making improvements, the absence of a shared understanding hampers effective collaborations between functions in efforts driving innovation forward. This lack of clarity contributes to confusion among employees regarding their work in the field of innovation.

The company possesses a highly competent workforce with extensive knowledge and a strong drive to innovate. While innovation is primarily influenced by external factors, the internal motivation and eagerness to change and innovate among the employees play an important role. This internal drive serves as an asset in the company's overall innovation efforts. Furthermore, they understand the importance of having and maintaining competence to foster innovation and to stay competitive in the market, which is reflected through the tendency of people staying within the company.

Although having a lot of internal competence, the company, as mentioned earlier, has challenges in managing resources, including an absence of overview of competence and support to allocate resources. There is a recognized need for implementing a tool or support system to manage resources effectively, specifically to identify gaps and address existing difficulties. Recognizing the importance of resource management, the company acknowledges that it is the responsibility of management to provide resources based on proven needs or identified gaps. To maximize utilization of competence, they emphasize the importance of aligning individuals with their respective strengths and placing them in appropriate positions. Overall, the company lacks a comprehensive overview of its resources, highlighting the importance of addressing this issue to enhance resource management capabilities.

The company faces challenges due to a lack of structure and time allocated for utilizing insights from follow-ups and implementing improvements. Although there are common ways of planning for projects, there are no shared structures to support follow-up during projects. There are meetings held to discuss improvements, but the frequency and consistency vary throughout the company. Furthermore, top management is receptive to improvement suggestions, but there is currently no formal and structured process in place for submitting such proposals. This point to the need of developing supporting structures and allocating resources to facilitate follow-up and improvement.

As brought up in relation to *Context of the Organization*, the company faces challenges with knowledge sharing and internal collaborations. The company lacks a structured approach to knowledge sharing and the knowledge sharing mostly relies on informal contexts and individuals' efforts. Although teams are organized based on competence, there is a lack of support to facilitate interdisciplinary collaborations. There is a report database for projects, serving as a valuable repository of lessons learned, goal analysis, execution plans, and improvement suggestions. However, there is currently no formalized process for utilizing the database, as individuals must independently search for relevant information.

The company currently faces a lack of structured processes for idea management, which has resulted in issues such as prematurely starting the development stage without a solid foundation. Recognizing this

issue, the company is actively working on establishing mechanisms to enhance the progression of ideas and determine their maturity level to provide the necessary support. They acknowledge that the absence of effective structures and methods for managing ideas in the early stages of development impedes the innovation process. Moreover, there is a lack of formal structures for idea generation and submission, with most ideas being informally shared during informal gatherings.

In relation to managing ideas and advancing them, the informants express the importance of access to testing and prototyping equipment. The company possesses advanced equipment that enables them to do testing and prototyping and relies on it in their development. However, there are limitations in accessing the provided equipment, leading to bottlenecks in the prototyping and testing process. Additionally, there is a risk of equipment breakage, which further hampers the company's ability to utilize these resources effectively. They recognize that ensuring reliable and sufficient access to testing and prototyping equipment is essential to support innovation and minimize disruptions in the development and validation of ideas.

The insights emphasize the prevailing lack of time and opportunity for innovation and exploration within the organization, as the primary focus is on efficiency. It is emphasized that both efficiency and innovation are crucial for future relevance. To enable a balance between the two and address current challenges, it is acknowledged that it is essential to establish a common view of direction, supporting structures within the organization, as well as allocate sufficient time for exploration and creativity.

4.1.5 Operation

Projects within the company are characterized by a low risk of failure due to requirements of proven feasibility and a robust business case before initiation and that projects are typically revised and adjusted, addressing challenges and changed circumstances, instead of being canceled. During project evaluation and selection, opportunities and potential rewards are carefully considered. To ensure successful project outcomes, the company recognizes the importance of establishing a solid foundation with clear goals and scope, as well as adequate resource planning and continuous follow-up. However, there is a lack of support or common framework for project follow-up, planning, and decision-making to ensure consistency.

The company also faces challenges in idea management. The absence of structured processes has resulted in premature development stages and a lack of a solid foundation for projects. There is a need for effective structures and methods to manage ideas in the early stages of development, as they are crucial for the innovation process. Additionally, the lack of formal structures for idea generation and submission has led to sharing of ideas in informal contexts. Although informal idea sharing is not an issue, however, it was expressed that resources and support is required to make room for and facilitate further development of ideas. Implementing a formalized idea management processes could enhance timeline alignment in activities and help develop ideas for successful innovation outcomes.

The increased focus on customer needs has supported the company in its innovation efforts. However, the company faces a risk of hindering new business opportunities by not exploring beyond its current

value chains. Their new ideas primarily revolve around current customers, potentially overlooking future customers and new business prospects. While maintaining a strong customer focus reduces the risk of failures, it may also impede innovation through limiting identification of opportunities.

As mentioned in *Support*, the availability of testing and prototyping equipment is crucial for advancing ideas within the company. However, there are limitations in accessing the provided equipment, leading to bottlenecks in the prototyping and testing process. Additionally, there is a risk of equipment breakage, which further hampers the company's ability to utilize these resources effectively. Finding ways to solve these challenges is expressed as essential in order to support validation of ideas and concepts.

The company follows a clear process model for development that includes planned deliverables, contributing to the progression of projects. However, while the stage-gate process effectively outlines the necessary deliverables, informants mentioned there is not clear guidance in ways of working to help achieve them. Although the company values freedom and flexibility in their approach, fostering creativity through diverse group constellations, informants expressed the need for tools and methods in ways of working that can enhance project outcomes without compromising flexibility. Additionally, there is a recognized need to prioritize learning from failures and proactively address bottlenecks to improve project performance and overall innovation efforts.

While the company's process is consistent for both incremental and radical innovation, it primarily aligns with the needs of incremental initiatives. The stage-gate process can be adjusted to accommodate radical innovation projects, but the company's core focus and expertise lie in incremental innovation. However, the informants acknowledge the significance of both types of innovation, and some highlight the need to develop the process to better support radical innovation. Thus, the process seems to be tailored to best support incremental innovation.

4.1.6 Performance Evaluation

As previously mentioned, the organization lacks a common structure for project follow-up, resulting in inconsistencies across product areas. While initial project planning includes risk analysis and goal setting, continuous follow-up during projects was recognized as a weakness. This highlights the need for a shared foundation to guide project planning, evaluation, and decision-making. Furthermore, the company faces challenges in utilizing follow-up insights and implementing improvements due to a lack of structure and allocated time. To address this, a new approach involving summarized conclusions is being tested for project evaluations.

To track innovation progress, the company uses key performance indicators (KPIs), focusing on patents and new product launches as indicators of success. This highlights their emphasis on financial and output related KPIs to measure innovation value. The evaluation of new products is primarily carried out by the product organization, with sales performance serving as a key criterion.

4.1.7 Improvement

The company faces challenges due to a lack of structure and time allocated for utilizing insights from follow-ups and implementing improvements. Top management is receptive to improvement suggestions, but there is currently no formal and structured process in place for submitting such proposals. While some meetings are held to discuss improvements, the frequency and consistency vary throughout the company.

4.1.8 Other

The company recognizes that achieving successful innovation requires finding the right balance between freedom and boundaries. While too many boundaries can impede creativity, excessive freedom may lead to a lack of focus or value. Harnessing creativity within a structured framework, was acknowledged to support innovation. In the early stages of ideation, it is important to have loose boundaries to encourage exploration, at the same time having a clear focus and well-defined structures can support progress. However, there is currently a lack of balance, resulting in room for creativity at the same time as projects are being initiated too early. Challenges arise when projects are initiated without clear initial conditions, leading to difficulties in aligning timelines.

4.1.9 Summary

The research findings are structured to align with the standard where each section of the findings corresponds to a specific element of the standard. The visual representation of this alignment can be observed in Figure 4.1, which serves as a visual guide, illustrating the interconnectedness and distribution of the findings. Additionally, for a quick reference and ease of navigation, Table 4.1 outlines the corresponding concepts associated with each number.

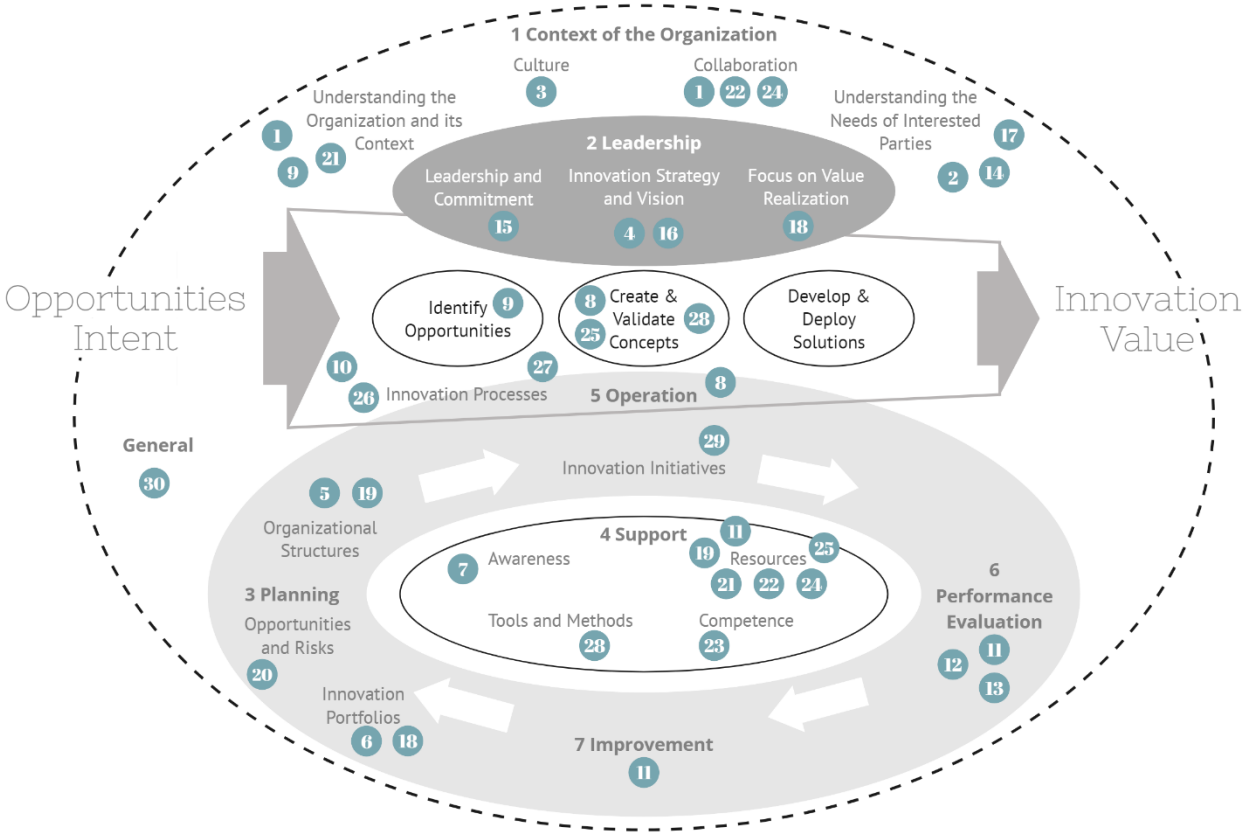


Figure 4.1. Concept distribution Company A according to the ISO 56002 standard, based on an adoption of ISO 56002:2019.

Table 4.1. Concepts with number and description Company A

#	Concept description	Related Element(s)
1	The company has external collaborations and ways to capture external trends and drivers but no formal and structured process to utilize insights from these	(1) Context of the Organization
2	It is important for the company to be close to customers and understand the end customer needs to establish a competitive advantage	(1) Context of the Organization
3	There is no failure culture and the company do not have structures and support for learning from failures	(1) Context of the Organization
4	There is adaptability in strategic direction if there are new insights, which could support in mitigating risks and uncertainties	(2) Leadership
5	When radical innovation is decided to be established in the company new organizational structures are developed to handle it	(3) Planning
6	The company started to consider the totality of projects and activities which support consistency	(3) Planning
7	There is no common language for innovation, although it is needed	(4) Support

#	Concept description	Related Element(s)
8	Projects are revised rather than cancelled and there is a low risk of projects failing as a solid business case is required prior to start	(5) Operation
9	The company does not explore outside of current value chains, potentially leading to overlooking new business opportunities	(1) Context of the Organization (5) Operation
10	The same process is employed for both types of innovation, but it is more adapted to incremental innovation	(5) Operation
11	The company lacks structure and time to utilize insights from follow up and to make improvements	(4) Support (6) Performance Evaluation (7) Improvement
12	There is a focus on financial and output related KPIs	(6) Performance Evaluation
13	The follow-up of projects differs throughout the organization and there is no common structure for it	(6) Performance Evaluation
14	Sustainability is an important driver for the company since it is crucial for survival and driven by both internal and external needs	(1) Context of the Organization
15	The management's responsibility is to provide direction and to facilitate innovation by providing support	(2) Leadership
16	There is a clear overall strategy formed and communicated top-down that provides direction for innovation and is followed up continuously	(2) Leadership
17	There is a total value approach (higher quality resulting in better performance) that could be a competitive advantage	(1) Context of the Organization
18	The company realize the importance to balance short term and long-term perspectives to ensure continuous value and relevance	(2) Leadership (3) Planning
19	There is a lack of time provided for innovation/exploration, the focus is on efficiency	(3) Planning (4) Support
20	Risk and uncertainties related to external conditions (e.g., price and access to raw material) could be handled by keeping informed and being flexible (e.g., having alternative suppliers)	(3) Planning
21	There is a lack of support to manage resources including overview of competence and planning/allocation	(1) Context of the Organization (4) Support

#	Concept description	Related Element(s)
22	The company lacks structure for knowledge sharing, it is mostly done in informal contexts	(1) Context of the Organization (4) Support
23	The company have very competent people with a lot of tacit knowledge and will to innovate	(4) Support
24	With a matrix organizational structure, it's important to facilitate collaboration and knowledge sharing between product areas	(1) Context of the Organization (4) Support
25	Availability to testing and prototyping equipment, which is crucial, but there is limited access and risk of breakage of equipment	(5) Operation
26	Clear process model for development with planned deliverables but need support in stages prior to development, clearing bottlenecks and ways of working	(5) Operation
27	Freedom in ways of working/adaptable project process and having different group constellations spur creativity	(5) Operation
28	The company lacks structures for idea management	(4) Support (5) Operation
29	The company expresses the importance to establish a foundation with clear scope and goals prior to projects and follow up on them	(5) Operation
30	Innovation requires a balance between freedom and boundaries. Boundaries could hinder creativity and freedom could result in misalignment to focus (lack of correspondence) or not adding value (invention rather than innovation)	-

4.2 Company B

Company B is an established company within the metal industry that specializes in providing innovative solutions and advanced technologies for different industries like the automotive, battery and steel industries. The company is also world leading within its area. In recent years, the company has made significant efforts to improve its innovation management practices. One aspect of this improvement involves a shift in their innovation approach from an inside-out to an outside-in perspective, focusing on addressing customer needs rather than imposing solutions based on internal beliefs. By aligning their innovation efforts with customer needs, the company aims to enhance their ability to create value and meet market demands more effectively. Furthermore, the company recognizes the many opportunities that the sustainability transition within the industry presents for innovation. Similar to Company A, sustainability is a key driver of innovation for Company B, and they are actively committed to playing a leading role in this transformative shift. This commitment is reflected in their involvement in projects that bring together various stakeholders from the industry. By participating in these projects, the company demonstrates its dedication to contributing to the sustainability transition and staying at the forefront of innovation in the field. The company operates its innovation activities through a central R&D unit that supports and collaborates on innovation efforts with the different business units. The innovation efforts are supported by top management and functions such as HR.

4.2.1 Context of the Organization

It is recognized that innovation can bring value both externally and internally, but it requires distinct approaches and considerations for each. The company has undergone a transition from an inside-out approach to an outside-in approach with a more customer-focused and solution-based approach, emphasizing the importance of understanding customer needs and becoming their preferred choice for value creation. However, there is currently a lack of common understanding regarding innovation and customer needs, resulting in a gap between development and customer requirements leading to a lack of market understanding. To address this, the company continuously monitors and gathers information about external trends including market trends, customer needs, megatrends, and specific technical trends at various levels within the organization.

In terms of external insights, there is a lack of structure to disseminate knowledge about the external context from sales to other parts of the organization. Although sales are responsible for monitoring the external environment, capturing all necessary perspectives can be challenging. The initial stages of understanding customer needs are sometimes overlooked, leading to missed opportunities. Furthermore, the company exhibits a slow pace in adapting and learning from others in terms of innovation practices, possibly due to hierarchical structures and bureaucratic processes that hinder agility and responsiveness.

The lack of communication between sales and product managers hinders product understanding, while the absence of structures and support inhibits collaboration between functions, units, and stages of development. The informants highlight the importance of bridging gaps between units such as sales,

product ownership, and R&D during major changes, and integrating diverse perspectives from different functions along with input and ideas from employees, since that enriches the innovation process. It is further brought up that collaboration between central R&D and other units, facilitated by shared understanding and common approaches, drives innovation.

There is a lack of structure, prioritization, and focus, resulting in a fragmented approach to innovation. Employees possess a strong drive to innovate and have significant knowledge and expertise and consultants are hired where the company lacks knowledge. However, there is a need to recognize the value of knowledge sharing and establish structures to support it. Currently, knowledge sharing is predominantly ad hoc and informal, leading to isolated knowledge and individual expertise. There are also challenges related to limited time and busy schedules that hinder effective knowledge sharing throughout the organization. However, knowledge exchange occurs between managers.

The company places a strong emphasis on cultivating a culture and work environment that fosters innovation and supports the retention of employees. Inclusion is actively promoted to create a safe and open organizational culture where individuals feel comfortable sharing their ideas. Although the culture allows for creativity, exploration and failure, there are no structures for learning from failures.

4.2.2 Leadership

The company has a well-developed business strategy that is collectively formed and includes prioritized areas for innovation. Even though there is no separate innovation vision or strategy, the existing business strategy provides clarity on the prioritized areas for the next five years. Top management holds responsibility for driving and validating the strategy, but the strategy development involves multiple functions and expertise. However, there is a challenge in effectively communicating this strategy from top management down to the rest of the organization. The connection between the strategy and innovation is therefore not well communicated and tends to be forgotten over time.

Top management demonstrates commitment to innovation through ambitious targets, investments, and support in terms of prioritization, training, and resource allocation. A shared direction can facilitate integration between functions and units, guide resource allocation and decision-making, and validate the work of R&D. Top management plays a key role in fostering innovation and driving improvements within the company. They have shown improvement in their ability to identify and implement necessary changes. However, it is important for management to understand that individuals may struggle with accepting changes, and guidance should be provided to support them through the transition. Additionally, top management demonstrates a willingness to take calculated risks if there is a solid business case, particularly in R&D, driven by a recognition of the need to explore new opportunities despite the inherent uncertainty and risk involved.

The company demonstrates a well-established process and structure for continuous improvement and development, as well as for highly radical opportunities involving external stakeholders. However, they face challenges in managing innovation that falls between these two categories. The focus is primarily on addressing near-future challenges, although efforts are made to include long-term perspectives.

Continuous improvements and new product development are treated differently, with the latter being viewed as more strategic and long-term, involving multiple functions.

4.2.3 Planning

The company recognizes the importance of having a shared direction to facilitate integration, resource allocation, and decision-making. However, there is a lack of structure and prioritization. The company also faces challenges in evaluating and selecting the most suitable opportunities amidst the increasing number of options brought about by electrification. While the industrial transition has positioned the company favourably, decision-making regarding which opportunities to pursue remains a complex task. The potential to create societal value, particularly in the context of sustainability, serves as a driving force for innovation.

Flexibility and agility are valued, allowing activities to adapt based on changing circumstances. Embracing agility, with a focus on iterations and prototyping, is seen as beneficial for driving innovation. It is essential to strike a balance between implementing structures and fostering a creative culture to promote flexibility. There is a need to continuously work towards becoming more agile which they recognize, although they further recognize that it is more challenging for an established company.

4.2.4 Support

To foster innovation, it is crucial for management to provide support, guidance, and create an environment where obstacles are removed. At the same time, it is important to strike a balance between having structures in place and fostering a creative culture that allows for flexibility and unscheduled activities. While it can be challenging for established companies to be agile, informants highlight as mentioned above, that the company should actively work towards becoming more flexible and adaptable to meet the evolving demands of the market and drive innovation.

The company benefits from an internal drive within its employees to innovate. However, there is a need to improve the sharing of individual knowledge, as it currently relies on ad hoc and informal methods, leading to isolated knowledge and individual expertise. Knowledge sharing primarily occurs between managers, but there is a wealth of knowledge and high competence among employees that is not effectively utilized, documented, or shared. This lack of sharing poses risks, such as the potential loss of knowledge from aging experts. Difficulties in sharing knowledge stem from the absence of structured mechanisms and the busy schedules of employees.

The company faces challenges related to the lack of a clear innovation vision and strategy, as well as a common language among top management and therefore also the rest of the organization. The absence of a shared understanding of innovation and customer needs hampers their ability to foster innovation effectively. Therefore, it is a recognized need to establish a common understanding and improve communication regarding the definition of innovation or transformative initiatives. Ongoing efforts are being made to create alignment and enhance clarity in these areas.

As mentioned before, the company's culture and work environment are supportive of innovation and employee retention. They actively promote inclusion to foster an organizational culture where individuals feel safe to express their ideas and thoughts freely. Retaining talent is considered crucial and the organizational culture encourages creativity, exploration, and the acceptance of failure as part of the innovation process. However, there is a lack of structured mechanisms for learning from failures. Although the organization values creativity and enables it through its culture, they recognize a need to analyse failures, extract lessons learned, and identify areas for improvement in a more structured manner. The ability to adapt and recover quickly from failures is essential for navigating significant changes, and prioritizing the application of lessons learned is crucial in this regard.

4.2.5 Operation

A shared direction within the company has the potential to facilitate integration between different functions and units, as well as support resource allocation, prioritization, and decision-making processes. It can also provide legitimacy to the work of the R&D. However, the company currently faces challenges stemming from a lack of structure, prioritization, and focus.

Having a flexible and adaptable framework for activities is crucial to navigate unexpected challenges and accommodate varying project scales. It allows the company to effectively respond and adjust when things do not go according to plan. This flexibility allows for adaptability and enables the company to make necessary adjustments to achieve desired outcomes. Additionally, embracing agility through iterative processes and prototyping can greatly benefit innovation efforts. There is a need to continuously work towards becoming more agile.

The company recognizes the distinction between continuous innovation and new product development (NPD), although it is not clearly expressed or emphasized. Continuous improvements and NPD are treated differently, with NPD being seen as more long-term and strategic, involving multiple functions. Continuous innovation is typically managed within the line organization, while higher degrees of innovation are handled through the NPD process. The innovation process can vary across different business units and is influenced by the type of innovation, whether it involves smaller changes or the development of entirely new products. The company demonstrates expertise in continuous innovation and large-scale shifts, including collaboration with external stakeholders, but faces challenges when it comes to innovation that falls in between these two, suggesting room for improvement in this area.

4.2.6 Performance Evaluation

Within the company, top management continuously follows up on innovation activities to support progress. However, follow-up and evaluation are primarily conducted at the top management level, with limited emphasis on lower levels of the organization. At the project and team level, some meetings for providing suggestions on improvements are held, but the evaluation process is mostly informal and relies

on conversations rather than structured methods. There is a recognized need to establish support for analysing and learning from failures and successes in order to capture valuable insights and identify areas for improvement.

The company primarily relies on financial metrics, including profit, new product offerings, patents filed, and cost reductions, to follow up on its innovation efforts. While some informants emphasize the need to go beyond financial metrics and explore additional indicators that can provide a more comprehensive understanding of innovation performance, there is a recognized challenge in effectively measuring innovation due to its inherent complexity and intangibility, compared to traditional financial metrics.

4.2.7 Improvement

As brought up in *Leadership*, top management plays a key role in facilitating innovation and driving improvements within the organization. They are responsible for identifying opportunities for innovation and implementing changes to enhance processes, products, and services. Informants expressed that individuals may find it challenging to accept and adapt to changes and therefore, effective management should provide guidance and support to navigate through these transitions.

Notably, the company has made progress in their ability to generate and work with improvements. Over time, management has developed better practices and approaches for initiating and managing change within the organization. This demonstrates a commitment to continuous improvement and a growing understanding of the importance of driving changes to enhance performance and innovation capabilities.

4.2.8 Other

The company utilizes a stage-gate model to guide the innovation process including idea evaluation, development, and commercialization. However, informants highlighted the need of establishing ways to assess ideas and measure progress in early-stage development. Management plays a key role in facilitating innovation by providing support, guidance, and necessary resources, while also removing obstacles that hinder progress. This reflects the shared opinion among informants that innovation requires finding a balance between creativity and flexibility, and structure. Structures offer direction, efficiency and support, while a creative culture encourages flexibility and exploration. While the company values creativity and promotes it through its organizational culture, several informants emphasized the need of support to further facilitate innovation.

4.2.9 Summary

The research findings are structured to align with the standard where each section of the findings corresponds to a specific element of the standard. The visual representation of this alignment can be observed in Figure 4.2, which serves as a visual guide, illustrating the interconnectedness and distribution of the findings. Additionally, for a quick reference and ease of navigation, Table 4.2 outlines the corresponding concepts associated with each number.

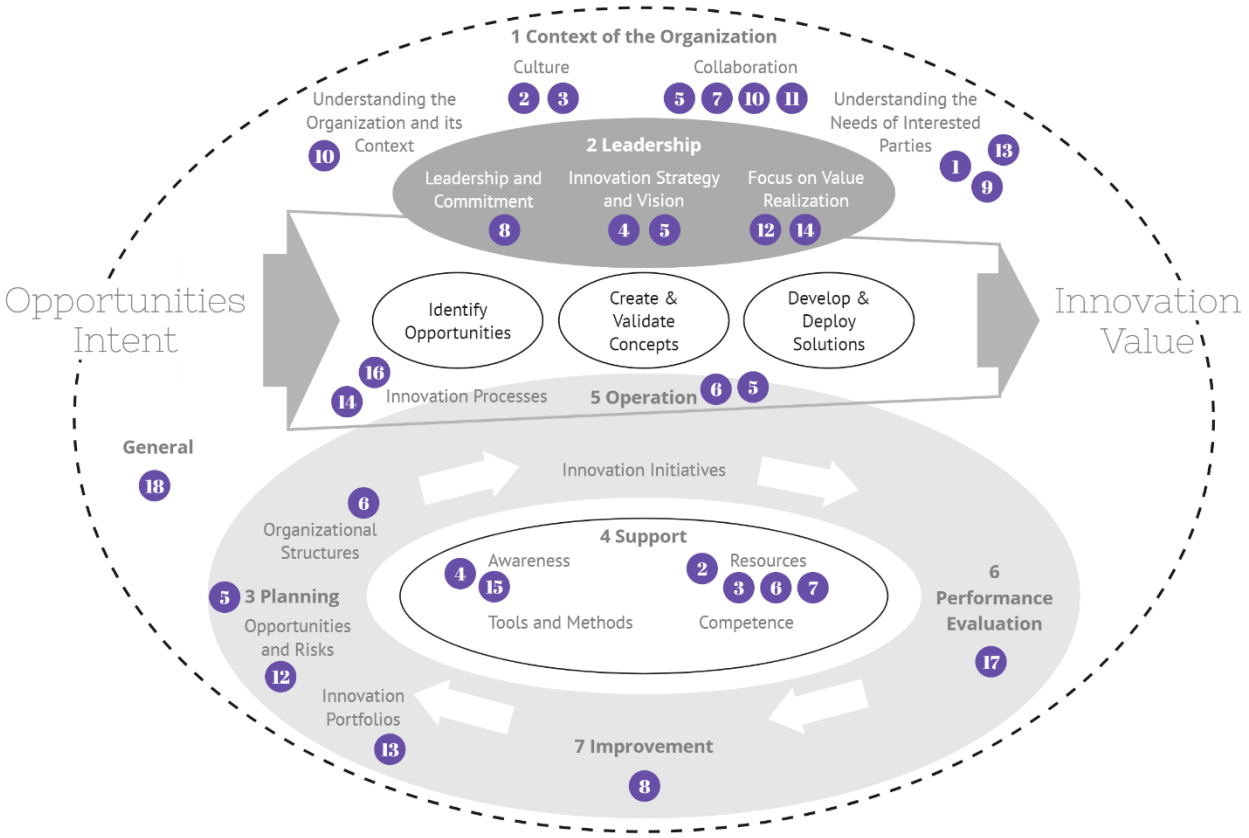


Figure 4.2. Concept distribution Company B according to the ISO 56002 standard, based on an adoption of ISO 56002:2019.

Table 4.2. Concepts with number and description Company B

#	Concept description	Related Element(s)
1	While transitioning from an inside-out approach to an outside-in approach, the company has made progress but still need to improve	(1) Context of the Organization
2	The culture and work environment support innovation and retainment of employees	(1) Context of the Organization (4) Support
3	The organizational culture allows for creativity, exploration and failure, but there are no structured ways of learning from failures	(1) Context of the Organization (4) Support
4	The company has a well-developed business strategy, formed collectively, that includes prioritized areas as focus for innovation, however it is not communicated well top-down	(2) Leadership (4) Support

#	Concept description	Related Element(s)
5	There is a recognized need for improved structure, prioritization, and focus in managing innovation since a shared direction enables function integration and decision-making prioritization	(1) Context of the Organization (2) Leadership (3) Planning (5) Operation
6	The company realizes that it is difficult for established companies to be agile, but that they should find ways to allow for flexibility as it is beneficial for innovation	(3) Planning (4) Support (5) Operation
7	There is a need to establish structures and provide resources to facilitate knowledge and idea sharing since it is done in an ad hoc and informal manner, leading to isolation of knowledge and individual expertise	(1) Context of the Organization (4) Support
8	The improvement efforts, which top management is responsible for, has undergone notable enhancements	(2) Leadership (7) Improvement
9	There is a recognized need of innovation aiming both internally and externally	(1) Context of the Organization
10	Despite ongoing monitoring of the external context, the company faces challenges in seizing opportunities due to a lack of knowledge and structured mechanisms to collect and share external insights	(1) Context of the Organization
11	To facilitate effective innovation and collaboration, the company need to establish structures and support systems that bridge gaps between different functions, enabling the exchange of diverse perspectives and fostering a common understanding	(1) Context of the Organization
12	Top management is willing to take calculated risks supported by a solid business case, especially in R&D, which aims to explore new opportunities despite high uncertainty and risk	(2) Leadership (3) Planning
13	There is a high potential and many opportunities with the sustainability transition within the industry, but it also comes with challenges related to deciding which opportunities to proceed with	(1) Context of the Organization (3) Planning
14	The company has a well working process and structure for continuous improvement and development and for highly radical opportunities that involve many external stakeholders, but have challenges with innovation that fall between these two	(2) Leadership (5) Operation
15	The company lack a common understanding and language for innovation at all levels, which hinder innovation	(4) Support

#	Concept description	Related Element(s)
16	The degree of innovation is not explicitly defined by the company, but incremental and radical innovation are handled differently during development	(5) Operation
17	There is continuous follow-up on top management level but is informal and unstructured at the project and team level. This coupled with the complexity and intangibility of measuring innovation creates difficulties in the follow-up process	(6) Performance Evaluation
18	It is understood by the company that innovation requires balance between on one hand creativity and flexibility and on the other hand structure, where structure can provide direction and guidance to create value from creativity	-

5 ANALYSIS

The analysis of the results is based on three dimensions of systemicity: comprehensiveness, coherence, and correspondence. These dimensions are employed to examine the impact of systemicity on the innovation capabilities of the two companies.

The *comprehensiveness* dimension of the system assesses the extent to which the innovation capabilities of the two companies identified in the study encompass the seven elements related to the standard, Context of the Organization, Leadership, Planning, Support, Operation, Performance Evaluation and Improvement, by considering both the coverage (spread) of capabilities over the elements and the maturity level (depth) of the capabilities. The mapping of the innovation capabilities over the elements, depicted in Figure 4.1 for Company A and Figure 4.2 for Company B, highlight areas where the companies exhibit a higher coverage of capabilities, as well as areas where they may have gaps or limited capabilities. It is important to note that the results are influenced by the specific questions asked during the interviews, which may have influenced the scope of the analysis for comprehensiveness. In some cases, assessing the maturity level of elements involved a two-step process: first assessing the maturity level of the areas within each element and then calculating the mean value of the included areas to evaluate the maturity level of the element as a whole. This approach was employed when there were significant variations in maturity levels among the areas within an element. For elements where the findings indicated a clear maturity level, this two-step process was not employed. The maturity level of the capabilities within the elements can range between 1 and 3, where Level 1 indicates that the capability is either non-existent or established in an informal or ad hoc manner, meaning that the capability is managed in a deficient manner or lacks a clear definition. Level 2 indicate that the capability is established at a basic level, meaning it is partially, but not fully, realized. Lastly, Level 3 indicates that the capability is defined and established with a formal character, as well as actively managed. Figure 5.1 presents a visual representation of the system dimension comprehensiveness, which depicts a system composed of various elements or subsystems.

The *coherence* dimension of the system assessment evaluates the relationships between elements identified in the findings. These relationships were identified by examining the elements to which the concepts pertained, considering whether the elements encompassed partial or complete aspects of the concepts. Figure 4.1 and Figure 4.2 illustrate that several concepts are connected to two or more elements from the standard, indicating a level of coherence among these elements. The interconnections between elements exhibit different types of relationships. In some cases, there is a unidirectional relationship where one element is dependent on or influences another element. In other cases, there is a bidirectional relationship where elements are mutually dependent or influential. Additionally, there is a mediator influence where one element influences the relationship between two other elements. Lastly, there is one relationship where two elements are related but not clearly dependent on each other, indicating a simple connection. It's important to note that the interconnections analyzed are based on concepts that explicitly show these connections, and other potential connections within the data are not

considered. Figure 5.1 provides a visual representation of the system dimension coherence, where examples of interconnections between elements or subsystems are illustrated by the connecting lines.

The *correspondence* dimension of the system assessment evaluates the alignment between the companies' actions and their goals. To conduct this analysis, innovation performance objectives related to managing both incremental and radical innovation were utilized. This approach was chosen considering the aim of the research project within Metallic Materials and the shared aim of both companies to be leading companies in the industry's transition towards sustainability. The dimensions of correspondence are particularly relevant for analysis as the transition necessitates radical innovation, while established companies often prioritize incremental innovation. The five innovation performance objectives (IPOs) employed for analysis include time, risk-taking, flexibility, novelty, and cost. Figure 5.1 provides a visual representation of the system dimension correspondence, where the alignment of elements or subsystems towards the goal is illustrated using dashed lines.

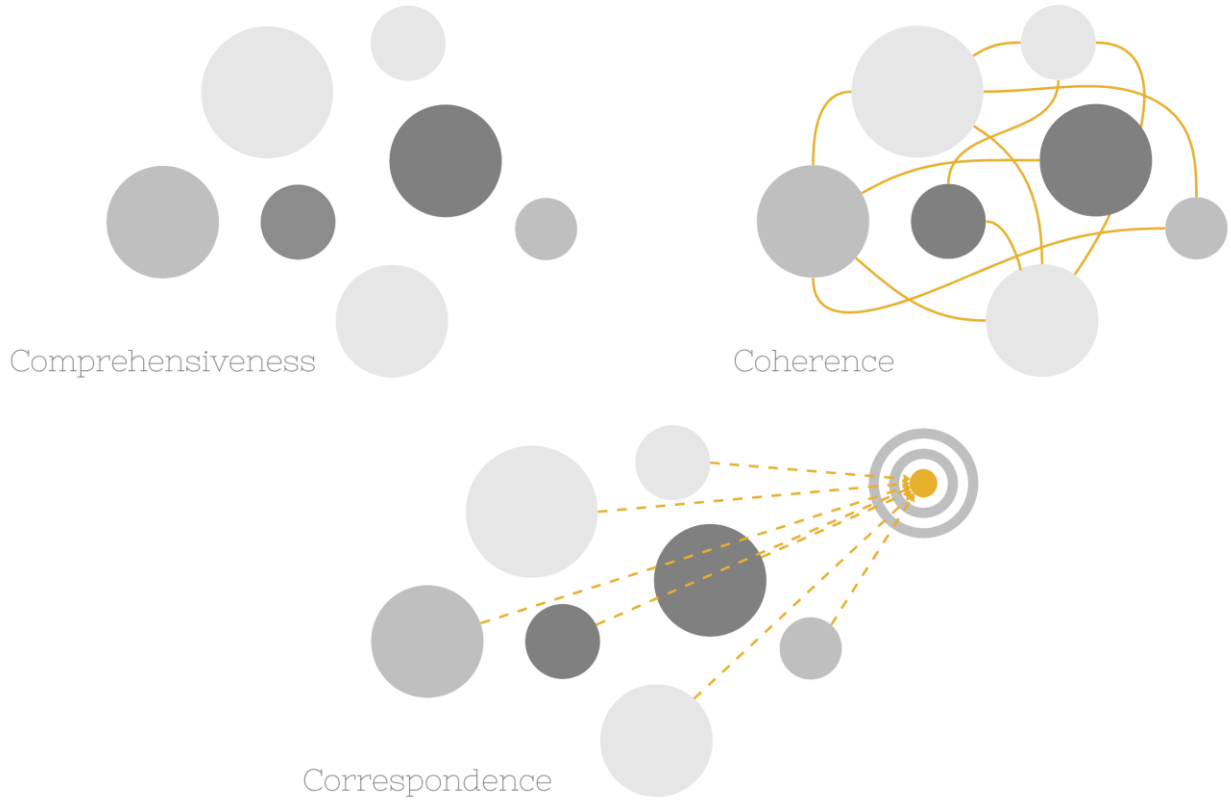


Figure 5.1. Illustration of the three system dimensions comprehensiveness, coherence, and correspondence.

The following sections of this chapter present the analysis of findings in relation to the three dimensions comprehensiveness, coherence and correspondence. First, company-specific analyses are presented to highlight the results specific to each case. This is followed by a cross-case analysis of both companies, to study similarities and differences, and to potentially identify patterns.

5.1 Company A

This chapter outlines the analysis of comprehensiveness, coherence and correspondence of Company A's innovation capabilities.

5.1.1 Comprehensiveness

The summarized maturity level of each element is presented in Figure 5.2. For a more detailed description of the comprehensiveness for each area in each element for Company A, see Table 5.1.

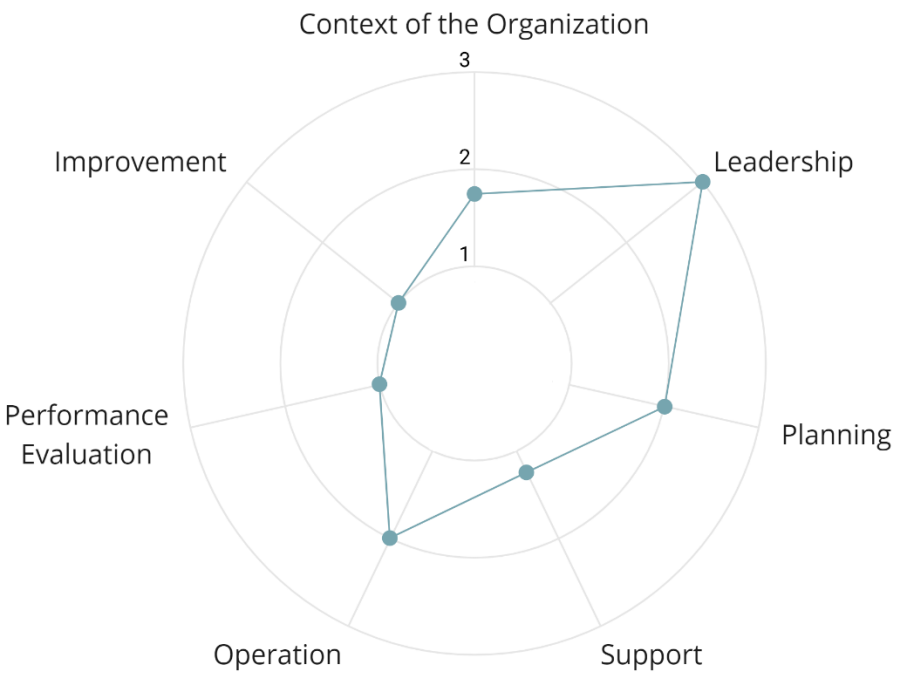


Figure 5.2. Radar chart summarizing the comprehensiveness by the total maturity level of each element of Company A. Note that the maturity level can range between 1 and 3, where Level 1: Nothing, informal, or ad hoc, Level 2: Managed at basic level, Level 3: Defined and managed.

Table 5.1. Concepts for Company A analysed with the system dimension comprehensiveness.

Element	**Area(s)**	**Maturity Level**	**Comment**	**Concepts**
Context of the Organization	Understanding the Needs of Interested Parties	3	• Actively striving to be close to customers • Prioritizing sustainability as it is driven by internal and external needs • Adopting a total value approach	2, 14, 17

Element	Area(s)	Maturity Level	Comment	Concepts
	Understanding the Organization and its Context	2	• Proactive approach to gathering external information but not effectively utilizing these • Not exploring beyond current value chains • Lack of support in managing resources and understanding internal capabilities, hindering effective resource allocation	1, 9, 21
	Collaboration	1	• None or informal knowledge sharing within the organization • Lack of collaboration between product areas	1, 22, 24
	Culture	1	• Lack of a failure culture	3
Leadership	Innovation Strategy and Vision	3	• Strategic adaptability in responding to new insights, mitigating risks, and embracing uncertainties • Presence of a formal overall strategy that guides innovation and is consistently followed up	4, 16
Planning	Opportunities and Risks	3	• Risks and uncertainties related to external conditions are actively addressed through staying informed and being flexible	20
	Innovation Portfolios	2	• Started considering the totality of projects and activities to achieve consistency	6
	Organizational Structures	1	• Limited time available for innovation and exploration activities	19

Element	Area(s)	Maturity Level	Comment	Concepts
Support	Competence	2	• Competent people with a lot of tacit knowledge and will to innovate, but not fully utilizing that knowledge	23
	Resources	1	• Lack of structure and time for follow-up and improvements • Lack of time for innovation and exploration, prioritizing efficiency • Insufficient support for resource management and knowledge sharing • Limited access and risk of equipment breakage in testing and prototyping	11, 19, 21, 22, 24, 25
	Tools and Methods	1	• Lacking structures for idea management	28
	Awareness	1	• No common language for innovation	7
Operation	Operational Planning and Control	3	• Proactive approach to risk management since projects undergo revision rather than cancellation	8
	Innovation Initiatives	2	• Important to establish a foundation with clear scope and goals prior to projects and follow up on them	29
	Innovation Processes	2	• Same process for both types of innovation, favouring incremental innovation • Clear development process but requires support for early stages and addressing bottlenecks • Some freedom in working methods and diverse group compositions (fostering creativity)	10, 26, 27

Element	Area(s)	Maturity Level	Comment	Concepts
	Create and Validate Concepts	2	• Low project failure risk due to solid business case requirement • Limited access and equipment breakage pose risks for testing and prototyping • Lacking structures for idea management	8, 25, 28
	Identify Opportunities	1	• Not exploring outside of current value chains which risks hindering new business opportunities	9
Performance Evaluation	Performance Evaluation	1	• Lack of structure and time for utilizing insights • Inconsistent follow-up of projects due to a lack of a common structure	11, 12, 13
Improvement	Improvement	1	• Lack of structure and time for making improvements	11
General	General	1	• No balance between freedom and boundaries	30

5.1.2 Coherence

The coherence between system elements identified from the findings for Company A is illustrated in Figure 5.3 below and described in more detail in the paragraphs following the figure. The arrow from one element to another implies dependability or influence and a simple connection through a line without an arrow implies that elements are related but without clear dependability or influence. Furthermore, the numerical values correspond to the concepts that indicate the relationship between elements.

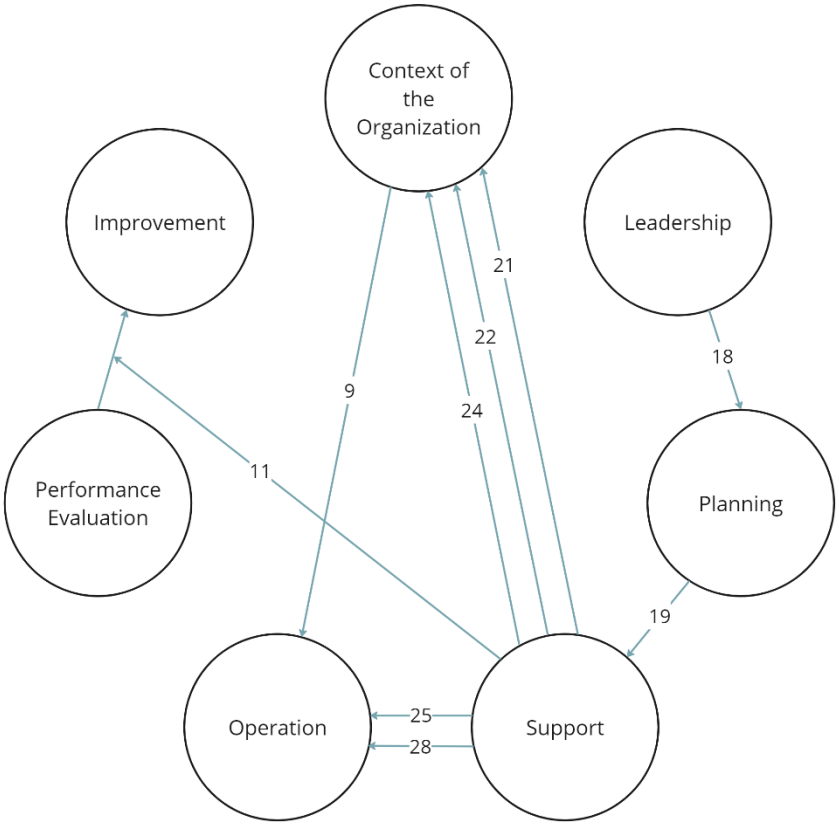


Figure 5.3. The coherence between the elements of Company A.
Explanation: arrow from one element to another implies dependability or influence, simple connection through line without arrow implies that elements are related but without clear dependability or influence. Additionally, an arrow from one element to the connection between two other elements indicates a mediator influence where the element influences the relationship between two other elements. The numerical values correspond to the concepts that indicate the relationship between elements.

The company's limited monitoring of external contexts beyond its current value chains poses a risk of overlooking potential business opportunities (Concept 9). This highlights the influence of external monitoring on the identification of business opportunities, suggesting a connection between the elements *Context of the Organization* and *Operation*. Specifically, it indicates a unidirectional relationship from the former to the latter element.

As indicated by Concept 11, Company A faces challenges in efficiently utilizing insights from follow-up processes to drive improvements due to a lack of structure and time. This suggests that there is a

relation between the evaluation and improvement processes, which can be influenced by the presence or absence of support. Consequently, it suggests that the standard element *Support* plays an intermediary role in shaping the unidirectional relationship between the standard elements *Performance Evaluation* and *Improvement*.

Concept 18 highlight the perceived importance of balancing short- and long-term perspectives to ensure continuous value and relevance. It was recognized from the informants that management is primarily responsible for achieving such a balance through taking it into consideration when forming the strategy, which relates to the standard element *Leadership*. The balance between different time horizons is also related to the element *Planning* as it should be considered in the management of innovation portfolios, looking at the totality of innovation initiatives (The International Organization for Standardization, 2019). Although not explicitly stated by the informants, it was expressed by some that innovation initiatives should and are presumed to align with the strategy. Furthermore, one of the product areas in the company applies a portfolio approach to manage activities and projects. Accordingly, a relationship between *Leadership* and *Planning* could be observed, where in this case, management of the innovation portfolio should reflect the goals outlined in the strategy.

Concept 19 address that there is much focus put on efficiency and that there is lack of time allocated for innovation, which suggest that the elements *Planning* and *Support* cohere. The planning process has an influential role in determining the level of support available for innovation activities. When efficiency is prioritized in planning, it may inadvertently limit the resources and time allocated for exploring new ideas and innovative initiatives. This lack of support in planning for sufficient time for innovation can hinder the company's ability to foster a culture of creativity and experimentation. Therefore, it becomes evident that planning directly influences the level of support provided for innovation-related efforts within the organization, which indicates a one-way relationship from *Planning* to *Support*.

Concepts 21, 22 and 24 indicate coherence between the standard elements *Support* and *Context of the Organization*. Insufficient support in managing resources, such as lack of support for planning and overview of competencies (Concept 21), hampers the efficient utilization of resources for innovation, leading to challenges in monitoring and addressing internal issues. Similarly, the absence of structured knowledge-sharing mechanisms, with most sharing occurring in informal contexts (Concept 22), limits the dissemination and utilization of valuable knowledge, thereby impacting internal collaboration. In a matrix organizational structure, support is essential to facilitate collaboration and knowledge sharing between functions and units (Concept 24) to foster effective communication and coordination across different parts of the organization. These three concepts imply that the element *Support* has an impact on the element *Context of the Organization*.

Concepts 25 and 28 suggest that the element *Operation* is dependent on the element *Support*. The availability and accessibility of testing and prototyping equipment play a critical role in progressing innovation within the company. However, the limited access to such equipment and the associated risk of breakage poses challenges in effectively utilizing them for innovation purposes. This lack of support

directly affects the innovation activities within the company by hindering the efficient execution of testing and prototyping activities. Similarly, the absence of structured processes for idea management further demonstrates the influence of support on innovation activities. Without proper frameworks in place, the company may struggle to effectively capture, evaluate, and implement innovative ideas.

To summarize the analysis of the findings in relation to coherence between the elements, it becomes evident that all elements have interconnections to other elements. Notably, the element *Support* emerges as particularly influential, with connections and influences on several other elements, underscoring its significance in driving the performance of these elements. Furthermore, a sequential relationship can be observed, with *Leadership* influencing *Planning*, which subsequently impacts *Support*, ultimately influencing various other elements within the system. These observations highlight the relationships and dependencies among the elements in the system.

5.1.3 Correspondence

The correspondence within the elements of Company A in relation to the Innovation Performance Objectives (IPOs) time, risk-taking, flexibility, novelty, and cost is outlined below and summarized in Table 5.2 in the end of this section.

Identified IPOs within the element *Context of the Organization* are **time** and **risk-taking.** Company A acknowledge the importance of sustainability for the long-term perspective due to it being a global trend, but they fail to utilize insights from other less obvious external trends and drivers. Therefore, there is partly a lack of correspondence in relation to the IPO time. The company's focus on being close to customers as well as not exploring outside of current value chains demonstrates a conservative approach that avoids taking risks and potentially limits new business opportunities. The absence of a failure culture and support for learning from failures suggests a reluctance to take risks and embrace radical innovation. Thus, there is a lack of correspondence in relation to the IPO objective risk-taking.

Identified IPOs within the element *Leadership* are **time, risk-taking, flexibility** and **cost.** Company A demonstrates an awareness of the importance of balancing short-term and long-term perspectives to ensure continuous value and relevance, which can support in achieving balance between incremental and radical innovation. Short-term goals might be more detailed and focused on efficiency and improvements that are feasible to implement close in time, whereas long-time perspectives can allow for exploring several opportunities and solutions more distant to existing offers. This awareness corresponds with the IPO time. Furthermore, the company demonstrates flexibility in adjusting their strategic direction based on new insights to mitigate risks and uncertainties. While this aligns with the IPO flexibility, there is a lack of correspondence with the objective of risk-taking. The company recognizes the crucial role of management in providing direction and support, including the allocation of resources, which corresponds with the cost objective. However, there are instances where support from management is lacking, indicating a partial lack of correspondence with the cost objective.

IPOs related to correspondence within the element *Planning* are **cost** and **time**. Due to a focus on efficiency, there is limited time allocated for exploration, resulting in a lack of resources provided to

radical innovation, which indicates a lack of correspondence with cost. However, when radical innovations have a proven business case, the company develops new organizational structures to support them, indicating a later-stage investment in more radical innovation. Therefore, the correspondence to the cost objective depends on the innovation's maturity level. The company acknowledge the significance of striking a balance between short-term and long-term perspectives for generating continuous value and staying relevant. As mentioned earlier, this awareness corresponds with the IPO time.

Identified IPOs within the element *Support* are **cost** and **flexibility**. The company's emphasis on efficiency over exploration results in a lack of allocation of resources, including time and money, to radical innovation. This shows a lack of correspondence with the IPO cost. The company's lack of structure and time to utilize insights from follow-up and make improvements reflects a lack of correspondence in terms of cost and flexibility. The company's desire to pursue more radical innovation may require changes in their ways of working, which necessitates the implementation of adjustments. However, the current lack of resources hinders the company's ability to carry out the necessary changes and in the long term, capitalize on these insights. Furthermore, the limited access to testing and prototyping equipment hinders the progress of radical innovation initiatives. This leads to a prioritization of incremental innovation projects that have a clear business case. As a result, more resources are allocated to incremental innovation efforts, which shows a lack of correspondence to the IPO cost.

IPOs related to correspondence within the element *Operation* are **flexibility**, **risk-taking** and **cost**. Company A recognizes the value of flexibility in ways of working and team constellations to facilitate radical innovation and nurture creativity. However, the company follows the same process for both incremental and radical innovation, although it is primarily adapted to incremental innovation. This implies a lack of flexibility in adapting the process to the requirements of radical innovation and therefore, there is a lack of correspondence with the IPO flexibility. Going into the development phase, Company A do not pursue projects without a proven business case, indicating willingness to take calculated risks. However, radical innovation sometimes exhibits high degrees of uncertainty during the development stage as well as later stages (Nilsson *et al.*, 2012). Therefore, the company could possibly miss out on opportunities, and thus, there is partially a lack of correspondence with the IPO risk-taking. The company, as mentioned, exhibit a lack of correspondence in relation to the cost objective as incremental innovation is prioritized over radical regarding access to testing and prototyping equipment.

IPOs related to correspondence within the element *Performance Evaluation* are **flexibility**, **cost** and **time,** and within the element *Improvement* **flexibility** and **cost**. The KPIs which the company uses for follow-up are primarily financial and output-related which may inadvertently impede radical innovation since radical innovation often requires a longer time perspective to generate significant value. Therefore, the reliance on KPIs that prioritize short-term financial and output metrics may not adequately capture the potential impact and value creation of radical innovation. This misalignment

results in a lack of correspondence with IPOs flexibility and time. The company's lack of structure and time to utilize insights from follow-up and make improvements reflects a lack of correspondence within the two elements in terms of cost and flexibility. The company's desire to pursue more radical innovation may require changes in their ways of working, which necessitates the implementation of changes. However, the current lack of investment of resources hinders the company's ability to implement improvements and in the long term, fully leverage valuable insights.

Table 5.2 below provides an overview of the company's correspondence status between the elements and the IPOs. An upward-pointing arrow indicates correspondence, a downward-pointing arrow indicates a lack of correspondence, an empty circle represents partial correspondence or an unclear status of correspondence, and the absence of a symbol indicates that the data does not provide information about correspondence.

Table 5.2. Summary of correspondence of Company A.
Explanation of symbols: arrow up = correspondence, arrow down = lack of correspondence, empty circle = partial correspondence or an unclear status of correspondence, nothing = data does not provide information about correspondence.

↑ Correspondence
↓ Lack of correspondence
○ Partial/unclear status of correspondence

	Time	Risk-taking	Flexibility	Novelty	Cost
Context of the Organization	○	↓			
Leadership	↑	↓	↑		○
Planning	↑				↓
Support			↓		○
Operation		↓	↓		↓
Performance Evaluation	↓		↓		↓
Improvement			↓		↓

5.2 Company B

This chapter outlines the analysis of comprehensiveness, coherence and correspondence of Company B's innovation capabilities.

5.2.1 Comprehensiveness

The summarized maturity level of each element is presented in Figure 5.4. For a more detailed description of the comprehensiveness for each area in each element for Company B, see Table 5.3.



Figure 5.4. Radar chart summarizing the comprehensiveness by the total maturity level of each element of Company B. Note that the maturity level can range between 1 and 3, where Level 1: Nothing, informal, or ad hoc, Level 2: Managed at basic level, Level 3: Defined and managed.

Table 5.3. Concepts for Company B analysed with the system dimension comprehensiveness.

Element	Area(s)	Maturity Level	Comment	Concepts
Context of the Organization	Culture	3	• Culture supporting innovation through allowing for exploration, creativity, embracing failure and retaining employees	2, 3
	Understanding the Organization and its Context	2	• Lack of knowledge and structured mechanisms to collect and share external insights	10

Element	Area(s)	Maturity Level	Comment	Concepts
	Understanding the Needs of Interested Parties	2	• Challenges in seizing external opportunities despite ongoing monitoring	1
	Collaboration	1	• Lack of leveraging external trends and drivers, requiring improved cross-functional collaboration and structured knowledge and idea sharing	7, 10, 11
Leadership	Focus on Value Realization	2	• Top management takes calculated risks in R&D to explore new opportunities despite uncertainty • Effective processes for continuous improvement and radical opportunities, but challenges with innovation that fall in between	12, 14
	Leadership and Commitment	2	• The improvement efforts, which top management is responsible for, has undergone notable enhancements	8
	Innovation Strategy and Vision	2	• Well-developed strategy, formed including several functions, but not well-communicated top-down creating a lack of focus	4
Planning	Opportunities and Risks	3	• Top management willing to take calculated risks, especially in R&D as their work is connected to many uncertainties	12
	Planning	1	• Lack of direction, including focus and prioritization, which hinders decision-making prioritization when planning activities etc	5

Element	Area(s)	Maturity Level	Comment	Concepts
Support	Awareness	1	• The company lacks effective communication and a shared understanding, hindering innovation efforts	4, 15
	Resources	1	• No structures for learning from failures • Flexibility is challenging for established companies but essential for innovation	2, 3, 6, 7
Operation	Innovation Processes	2	• Effective processes and structures for continuous improvements and radical innovations, but challenges with the in-between	14
	Operation	1	• Improvements needed in structures, priorities, and decision-making to integrate functions and enhance focus	5, 6
Performance Evaluation	Performance Evaluation	2	• The improvement efforts, which top management is responsible for, has undergone notable enhancements	17
Improvement	Improvement	2	• Follow-up is conducted, but informally at lower levels	8

5.2.2 Coherence

The coherence between system elements identified from the findings for Company B is illustrated in Figure 5.5 below and described in more detail in the paragraphs following the figure. The arrow from one element to another implies dependability or influence and a simple connection through a line without an arrow implies that elements are related but without clear dependability or influence. Furthermore, the numerical values correspond to the concepts that indicate the relationship between elements.

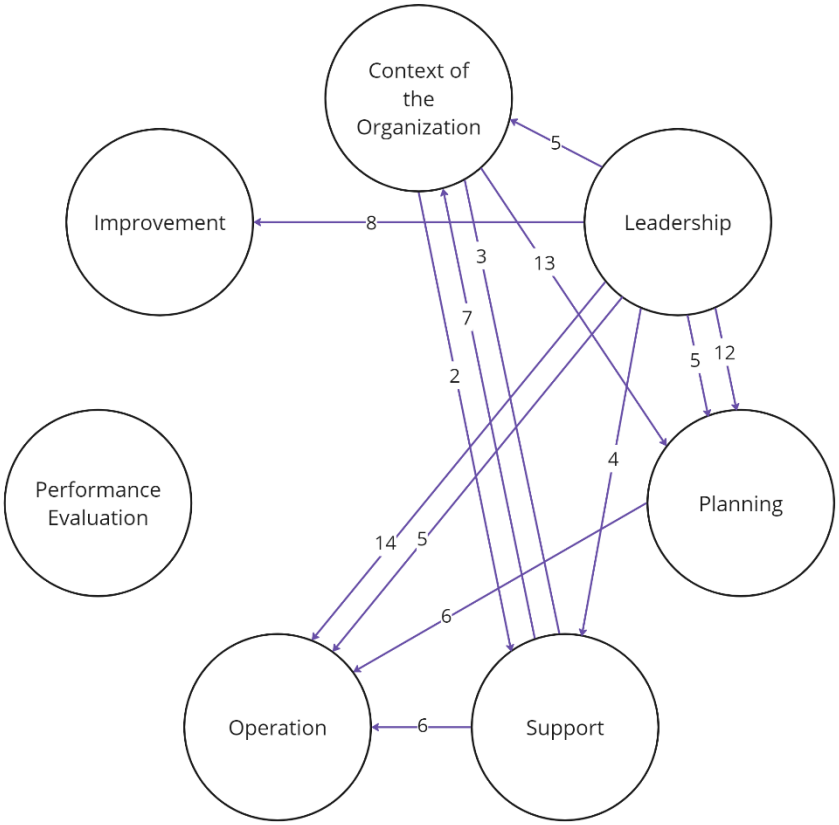


Figure 5.5. The coherence between the elements of Company B.
Explanation: arrow from one element to another implies dependability or influence, simple connection through line without arrow implies mutual dependability or influence. The numerical values correspond to the concepts that indicate the relationship between elements.

Concepts 2, 3 and 7 indicate coherence between the standard elements *Support* and *Context of the Organization.* The culture and work environment of Company B foster innovation and contribute to employee retention (Concept 2). This demonstrates the impact of the organizational context and culture on retaining valuable resources within the company and preserving knowledge. This indicates a one-way relationship between the elements *Context of the Organization* and *Support,* from the former to the latter. Furthermore, while the organizational culture of Company B encourages creativity, exploration, and accepting failure, there is a lack of structured mechanisms to learn from those failures (Concept 3). This relates to the element *Support*, which emphasizes the importance of leveraging lessons learned from both successful and unsuccessful innovation initiatives. Hence, there is a clear connection between the elements *Context of the Organization* and *Support*. Additionally, the informal structures

and resources to facilitate knowledge and idea sharing leads to isolation of knowledge and individual expertise (Concept 7), hence there is another unidirectional relationship between *Support* and *Context of the Organization*, from the former to the latter.

The organization's well-developed business strategy, formulated collectively, identifies prioritized areas for innovation (Concept 4). However, the lack of effective top-down communication impedes the alignment and coordination between leadership and support functions in leveraging the strategy, implying that the element *Support* is dependent on *Leadership*.

Concept 8 underscores the responsibility of top management in promoting innovation and driving improvements within the organization. Top management has made progress in this regard and is actively working towards identifying issues and implementing necessary improvements. This highlights a relationship between *Leadership* and *Improvement.*

Concept 14 brings up the company's well-working processes and structures for continuous improvement and development, as well as for highly radical opportunities involving external stakeholders, which is connected to the element *Leadership*. However, the challenges faced by the company in innovation that falls between these two categories highlight the need to bridge the gap between strategy and execution of innovation activities. This implies a relationship between *Leadership*, represented by the ability to establish processes and structures, and the operational aspect of innovation within the organization, related to the element *Operation.* Here, the element *Leadership* influences the element *Operation*.

The relationship between *Leadership* and *Planning* can be observed through top management's willingness to take calculated risks supported by a solid business case, particularly in R&D (Concept 12). This demonstrates the proactive leadership approach in fostering innovation through embracing uncertainty and risk. By making informed decisions and investing resources in exploring new opportunities, top management sets the direction and provides the necessary guidance for the planning function. The planning process can then align with the organization's strategic goals and objectives, incorporating the calculated risks identified by top management. Hence, *Leadership* influences *Planning.*

Concept 5 shows dependencies between multiple elements, from *Leadership* to *Context of the Organization, Planning* and *Operation*. The concept highlights that there is a recognized need for improved structure, prioritization, and focus in managing innovation since a shared direction enables function integration and decision-making prioritization. In *Leadership*, the strategic direction and vision is set for the organization's innovation efforts, which influence internal collaboration and integration between functions related to the element *Context of the Organization.* Furthermore, it has an impact on *Planning* in terms of setting priorities and objectives for innovation which can guide the planning process, for example in prioritization of resources. Lastly, it has an influence on the implementation and execution of innovation initiatives and thus, the element *Operation.*

The high potential and many opportunities with the sustainability transition within the industry presented by Concept 13 is included in *Context of the Organization*. This concept further explains that the transition comes with challenges related to deciding which opportunities to proceed with, which relates to the element *Planning*. Hence, a relationship between these two elements exists, from the former to the latter.

The company acknowledges that it is difficult for established companies to be agile, but that they should find ways to allow for flexibility as it is beneficial for innovation (Concept 6). The organizational structure is included in the element *Planning*, which is then related to *Operation* in terms of finding ways to introduce flexibility in their work structures. Another element influencing *Operation* is *Support*, in terms of providing the necessary resources and guidance to adapt to changing circumstances and embrace flexibility in work practices.

To summarize the analysis of the findings in relation to coherence between the elements, it becomes evident that most elements exhibit interconnections with other elements. However, the concepts do not suggest a specific relationship between *Performance Evaluation* and any other element. In order to establish a system perspective, it is important for all elements to have connections to one another. It is worth noting that the analysis of coherence only considered the visible relations derived from the mapping of concepts, without taking logical connections into account. Notably, the element of *Leadership* emerges as particularly influential, as it impacts several other elements while remaining unaffected itself. This underscores its role in driving the performance of these interconnected elements. Additionally,
the elements *Context of the Organization* and *Support* exhibit coherence with multiple other elements, indicating their interconnectedness and interdependence within the system. The findings suggest that these two elements rely on each other in various ways, creating a mutually dependent relationship. Additionally, the element *Operation* is influenced by and dependent on several other elements, signifying the numerous dependencies among elements necessary to effectively implement innovation initiatives.

5.2.3 Correspondence

The correspondence of the elements of Company B in relation to the Innovation Performance Objectives (IPOs) time, risk-taking, flexibility, novelty, and cost is outlined below and summarized in Table 5.4 in the end of this section.

IPOs related to correspondence within the element *Context of the Organization* are **risk-taking** and **time**. Company B fosters an organizational culture that encourages creativity, exploration, and acceptance of failure, creating an environment that facilitates radical innovation. However, the absence of structured mechanisms for learning from failures indicates a partial correspondence with the objective risk-taking. The company demonstrates awareness and recognition of the multitude of opportunities presented by the industry's sustainability transition. They have identified these opportunities and are conscious of the necessary time horizon and resource allocation. This level of awareness corresponds with the innovation performance objective of time, as it reflects a consideration of the long-term perspective and the investments required for sustainable innovation.

Highlighted IPOs within the element *Leadership* are **cost, risk-taking, novelty** and **time.** Top management's willingness to take calculated risks supported by a solid business case, particularly in R&D, demonstrates a partial correspondence with the IPO of risk-taking. Furthermore, in terms of novelty and cost, the willingness to take calculated risks and invest in R&D implies a desire to explore and introduce novel and radical ideas. Therefore, this also shows correspondence in relation to the IPOs novelty and cost. Top management's improved facilitation of innovation and continuous improvement indicates their investment in innovation, which corresponds to the IPO cost. The well-functioning process and structure for continuous improvement and development, as well as for highly radical opportunities involving external stakeholders, corresponds with the IPOs of time and novelty. However, there are challenges with innovation that falls between these two categories. The company may allocate fewer resources to radical innovation that is not driven by external forces, which may impact the cost objective, therefore it results in a lack of correspondence to cost.

All IPOs, **time**, **risk-taking**, **flexibility**, **novelty** and **cost** exhibit a relation to correspondence within the element *Planning*. As previously discussed, the company acknowledges and has identified the many opportunities that comes with the sustainability transition within the industry and is aware of the time horizon and resources required. This awareness corresponds with the IPO time. As also mentioned earlier, the company's top management exhibits a willingness to take calculated risks if there is a solid business case, especially in R&D, which demonstrates partial correspondence with the innovation performance objective of risk-taking. Additionally, the willingness to take risks and invest in R&D indicates a desire to explore and introduce novel and radical ideas, which corresponds to the objectives of novelty and cost. There is an awareness within the company that established companies such as themselves often face challenges in being agile, but that it is crucial for them to explore ways to foster flexibility as it holds significant benefits for innovation in general, and radical innovation in particular. Although demonstrating this awareness, the company to some extent struggles to enable flexibility, which implies that there is partial lack of correspondence with the IPO flexibility.

Identified IPOs within the element *Support* are **risk-taking** and **flexibility**. The organizational culture fosters creativity, exploration, and acceptance of failure, which indicate a willingness to take risks which corresponds to the IPO risk-taking. However, the lack of structured ways to learn from failures highlights a lack of correspondence with the same IPO risk-taking. Furthermore, the company acknowledges the challenges that established companies face in being flexible, which is crucial for driving radical innovation initiatives. This awareness corresponds with flexibility.

The IPOs related to correspondence within the element *Operation* are **flexibility, cost, time** and **novelty.** As mentioned before, the company's recognition of the challenges faced by established companies in terms of flexibility highlights the importance of this IPO for driving radical innovation initiatives. The awareness corresponds with the IPO flexibility. The processes and structure support continuous improvement and highly radical opportunities, aligning with the IPOs of time and novelty. However, there are challenges with innovation falling between these categories where fewer resources are allocated which may affect the cost objective, indicating a lack of correspondence in this area. The degree of innovation is not explicitly defined, but it is managed differently during the development process. This demonstrates correspondence to flexibility in handling both incremental and radical innovation initiatives.

Within the element *Performance Evaluation*, no specific findings were identified regarding the correspondence with the IPOs. The IPO related to correspondence within the element *Improvement* is **cost.** Top management is responsible for facilitating innovation and driving improvements within the organization, which has shown improvement over time. In relation to cost, the company recognizes the need to invest in initiatives that are suitable for radical innovation, which corresponds to IPO cost.

Table 5.4 below provides an overview of the company's correspondence status between the elements and the IPOs. An upward-pointing arrow indicates correspondence, a downward-pointing arrow indicates a lack of correspondence, an empty circle represents partial correspondence or an unclear status of correspondence, and the absence of a symbol indicates that the data does not provide information about correspondence.

Table 5.4. Summary of correspondence of Company B.
Explanation of symbols: arrow up = correspondence, arrow down = lack of correspondence, empty circle = partial correspondence or an unclear status of correspondence, nothing = data does not provide information about correspondence.

↑ Correspondence
↓ Lack of correspondence
◯ Partial/unclear status of correspondence

	Time	Risk-taking	Flexibility	Novelty	Cost
Context of the Organization	◯	◯			
Leadership	↑	◯		↑	◯
Planning	◯	◯	◯	↑	↑
Support		◯	◯		
Operation	↑		◯	↑	↓
Performance Evaluation					
Improvement					↑

5.3 Cross-Case Analysis

Comparison of the findings between the two companies for each system dimension are presented in the following chapters.

5.3.1 Comprehensiveness

As for the comprehensiveness of the innovation capabilities of Company A and Company B, Figure 5.6 reveals some distinctions between the two organizations. From the analysis of the innovation capabilities of both companies, it becomes evident that Company B showcases a more balanced distribution across its elements compared to Company A, which exhibits a more fragmented picture. Notably, Company A demonstrates a high maturity level in *Leadership*, while having relatively lower maturity levels in *Improvement* and *Performance Evaluation*. On the other hand, Company B maintains relatively balanced maturity levels across the elements but highlights lower levels within *Support* and *Operation*. These observations provide insights into the elements where the companies' innovation capabilities are either actively managed or not. Studying the two companies together in relation to the system dimension comprehensiveness provides an opportunity to gain insights into the similarities and differences between their innovation capabilities, which further can provide implications of patterns in how various innovation capabilities are managed.

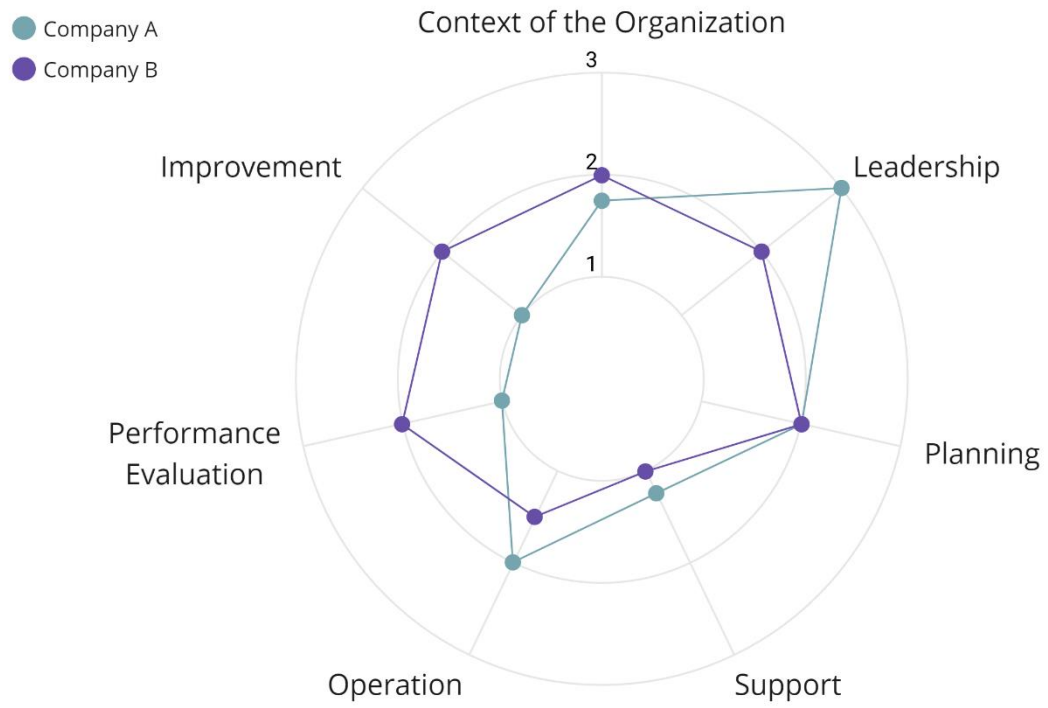


Figure 5.6. Comparison of the comprehensiveness between Company A and Company B. Note that the maturity level can range between 1 and 3, where Level 1: Nothing, informal, or ad hoc, Level 2: Managed at basic level, Level 3: Defined and managed.

5.3.2 Coherence

For the comparison of coherence between the capabilities of Company A and Company B, Figure 5.7 illustrates the interconnections between the elements in both organizations. It is evident that all elements have connections with other elements, showcasing their interdependence within the system. Notably, certain elements, such as *Leadership* and *Support*, exhibit a higher number of relationships and influences. These elements play influential roles, affecting other elements in the system. Specifically, *Planning*, *Context of the Organization*, and *Operation* are primarily influenced by other elements, while *Context of the Organization* shows a mutual dependency, particularly with the element *Support*. On the other hand, *Improvement* and *Performance Evaluation* stand out for their limited dependencies, suggesting a potential gap in terms of their relationships with other elements. Analyzing the two companies together in terms of the system dimension coherence offers an opportunity to explore the similarities and differences in the interconnections between the elements. This analysis helps identify elements that exhibit greater influence or dependency on other elements. Examining the interconnections can provide insights into the dynamics of the system and identify specific relationships that require attention and further exploration.

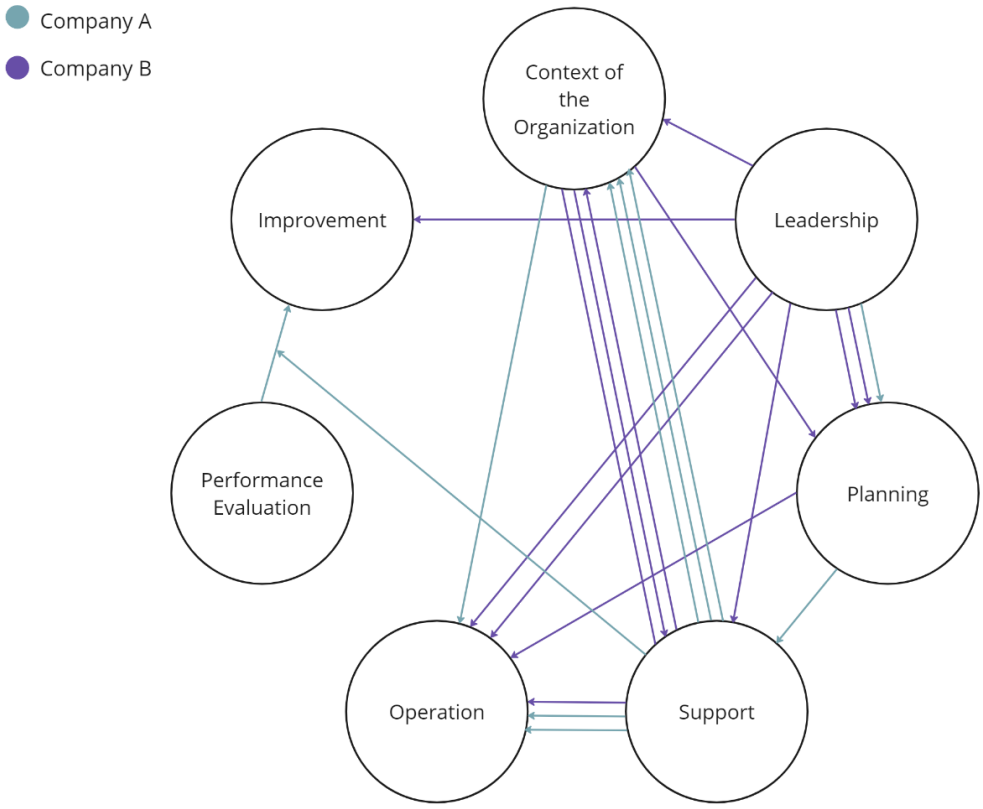


Figure 5.7. Comparison of the coherence between Company A and Company B. Explanation: arrow from one element to another implies dependability or influence, simple connection through line without arrow implies that elements are related but without clear dependability or influence. Additionally, an arrow from one element to the connection between two other elements indicates a mediator influence where the element influences the relationship between two other elements.

5.3.3 Correspondence

Table 5.5 presents a comparison of the correspondence between the innovation capabilities and the innovation performance objectives between Company A and Company B. The companies demonstrate many differences, indicating that they have different challenges and strengths in relation to correspondence. *Leadership*, *Planning*, and *Operation* emerge as significant elements, exhibiting multiple relationships with innovation performance objectives. The element that exhibits the greatest similarity between the two companies is *Context of the Organization*, as they both demonstrate similar correspondence with the same two IPOs. However, a distinct contrast arises in *Performance Evaluation*, where only Company A demonstrates any corresponding elements. Similarly, only Company B demonstrates correspondence with the IPO novelty.

Table 5.5. Comparison of the correspondence between Company A and Company B. Explanation of symbols: arrow up = correspondence, arrow down = lack of correspondence, empty circle = partial correspondence or an unclear status of correspondence, nothing = data does not provide information about correspondence.

Company A

Company B

	Time	**Risk-taking**	**Flexibility**	**Novelty**	**Cost**
Context of the Organization	○ ○	↓ ○			
Leadership	↑ ↑	↓ ○	↑	↑	○ ○
Planning	↑ ○	○	○	↑	↓ ↑
Support		○	↓ ○		○
Operation	↑	↓	↓ ○	↑	↓ ↓
Performance Evaluation	↓		↓		↓
Improvement			↓		↓ ↑

6 DISCUSSION

This chapter serves to address the research questions posed in this study and discuss the limitations of the study. Additionally, it presents a tentative framework aimed at analyzing system dynamics, identifying obstacles and opportunities within innovation management approaches.

6.1 RQ 1 - The Influence of Innovation Management Systemicity on Innovation Capabilities

The study has investigated the influence of innovation management systemicity on organizations' innovation capabilities, aiming to answer the following research question:

> RQ 1: How does innovation management systemicity influence an organization's innovation capabilities, where systemicity is defined by the three dimensions (1) comprehensiveness, (2) coherence, and (3) correspondence?

The first dimension, comprehensiveness of the system, was studied considering two aspects, spread of capabilities over the system elements, and depth or level of maturity of capabilities within each element. Looking at the spread of capabilities, the results indicate that the system elements considered in this study are relevant for the two companies' overall innovation capability. The data suggests that various aspects within these elements, or elements themselves, support successful innovation management. The study emphasizes the significance of direction and competent employees in initiating innovation, highlighting the importance of the elements *Leadership* and *Support* during these stages, serving as a foundation for fostering innovation within the organizations. However, the findings also imply that all elements are necessary to progress from input to output. When considering the depth, it remains unclear if a high maturity level is desirable for every element or parts of the elements.

The analysis of the results indicate that all studied elements are interconnected, highlighting their interdependence within the system. Furthermore, none of the companies explicitly adopt a systemic approach, yet implicitly acknowledge the importance of all the elements. The lack of support and awareness regarding interdependencies between elements seem to hamper the success of both companies in achieving desired outcomes, despite their individual capabilities within multiple elements. Recognizing the impact of coherence between different parts of the organization is important because decisions and actions made addressing one element will have an impact also elsewhere within the organization. For example, having competent individuals alone is insufficient if there is no support to utilize their competence. Similarly, having an ideation system is futile if insufficient time is allocated to it, leading to suboptimal outcomes, one of the effects of not having an holistic approach as emphasized in literature (Magnusson, Kristiansen and Bessant, 2022). Some relationships between elements are supported by multiple first-order concepts from the results, further emphasizing the coherence within the system. However, it is worth noting that the study reveals that there are limited connections to the elements *Performance Evaluation* and *Improvement*. Informants within both companies expressed

difficulties in evaluating and measuring innovation, indicating a potential gap in the system. It is important to address this gap and establish appropriate measurement mechanisms as prioritizing innovation requires understanding how to improve (Frishammar *et al.*, 2019). The overarching goal of the entire innovation system is to create value (O'Reilly and Tushman, 2004), and in order to enhance that value, it is necessary to identify areas for improvement.

With regard to the dimension of correspondence, the two studied companies exhibit divergent findings. These findings reflect the alignment between the companies' efforts and the goals established in this study. It is important to note that these goals are not explicitly formulated by the companies themselves but were established in this study based on literature with the aim to reflect the innovation performance objectives required for radical innovation, particularly in the context of large-scale shifts such as the sustainability transition. Furthermore, the results indicate multiple instances where the status of correspondence between capabilities within the elements and IPOs intersect. Consequently, drawing conclusions about the actions the two companies should undertake, or the order in which they should be pursued, becomes challenging. Therefore, it would be beneficial establish a prioritization of the IPOs in relation to each element, thereby facilitating decision-making.

Another observation in relation to the dimension correspondence in the two cases is the absence of alignment between the actions taken by the companies in some areas and their desired outcomes. Logically and as suggested by the analysis, not having capabilities correspond with intended goals seem to hamper the achievement of these goals. Consequently, it is important to be aware of how innovation capabilities align with the objectives of innovation. Simply following general directives without understanding their purpose may prove insufficient. However, this raises a discussion on whether a highly focused and rigid system can stifle the emergence of changes and variations, ultimately impeding innovation (Reinertsen and Shaeffer, 2005). In other disciplines where systemic approaches have been employed, the desired outcomes are easier to define and predict compared to the multitude of potential outcomes associated with innovation. This raises the question of how systemic an innovation management approach should be. Is a high degree of systemicity, including comprehensiveness, coherence, and correspondence, desirable? Arguably, as mentioned earlier, it may not be the case. There are findings in the study in relation to this, where for example Company A has developed a strong sustainability focus, primarily targeting a specific industry. Informants highlighted that this focus has constrained exploration in other areas where for example employees possess expertise or where there are potential business opportunities.

Another example pertains to Company B, where the informants associated innovation primarily with the development and delivery of products or offerings. However, they emphasized the importance of engaging in various forms of innovation, targeting both internal and external aspects, including innovation in ways of working. Structures shape behaviour (Eisenhardt, Furr and Bingham, 2010) and if innovation management practices exhibit high levels of systemicity, there is a potential risk of aligning innovation efforts with the existing system, thereby inhibiting risk-taking and flexibility. This limitation could potentially also extend to different forms of innovation, as those that align with the management

system are more likely to gain an advantage over those that do not. Consequently, there exists challenges with systems approaches as such approaches could potentially constrain the range of potential outcomes from innovation activities and implicitly favour incremental innovation. It is important to note that neither of the two companies explicitly adopts a systemic approach to managing innovation, and therefore it is difficult to draw conclusions about the effects of employing approaches where frameworks for innovation management systems are utilized. However, these observations highlight the possible challenges inherent in systemic approaches. Managing innovation entails risk-taking and being responsive to emerging information (Reinertsen and Shaeffer, 2005), thereby necessitating an innovation management system that enables exploration beyond existing knowledge, as well as adaptability of and variability within the system. A highly structured innovation management approach can impede necessary flexibility, while a lack of structure enables flexibility but may result in a failure to deliver value due to a lack of support for utilizing that flexibility (Eisenhardt, Furr and Bingham, 2010). Therefore, as emphasized in interviews conducted with both companies, striking a balance between flexibility and structure is essential, indicating the need for a balanced level of systemicity. For established companies, these challenges become particularly significant as they often prioritize efficiency over flexibility (Bhidé, 1999; Eisenhardt, Furr and Bingham, 2010). Consequently, they must adapt and revise their structures to prevent rigidity within the system. Clearly, additional research is required to further advance the knowledge regarding systemicity and innovation management systems.

As a final remark, the three dimensions applied to study the systemic properties and dynamics of the two companies' innovation management approaches: comprehensiveness, coherence and correspondence, have contributed with different perspectives to the analysis. The inclusion of these dimensions in the analysis allows for an extensive evaluation of the systemic nature of the innovation management approaches. It helps identify areas of strength and areas for improvement, providing a more holistic perspective on the overall system's potential for fostering innovation. The dimensions also reflect the definition of a system, that systems consist of set of elements that are interrelated and have a common function or purpose (Segatto, Inês Dallavalle de Pádua and Pinheiro Martinelli, 2013). Therefore, they are suggested suitable dimensions for capture the concept of systemicity.

6.2 RQ 2 - Hindrances and Opportunities for Managing Incremental and Radical Innovation within the same System

The study aimed to not only examine the influence of innovation management systemicity on organizations' innovation capabilities but also investigate the challenges and opportunities associated with managing both incremental and radical innovation with a systems approach by addressing the following question:

> RQ 2: What does systemicity indicate regarding hindrances and opportunities for managing incremental and radical innovation within the same system?

A systems perspective is important for understanding innovation management and the possibility of handling both radical and incremental innovation within the same system. When examining a system, it is important to consider its components, how they are interconnected and if these components collectively contribute to the system's function as outlined in both literature (Segatto, Inês Dallavalle de Pádua and Pinheiro Martinelli, 2013) and the discussion of the first research question. The question of whether both radical and incremental innovation can be managed within the same system moves closer to an answer when the function of the system in the context of innovation is clearly defined. Ultimately, it revolves around value creation (O'Reilly and Tushman, 2004).

In Company A and Company B's pursuits to become leaders in the sustainability transition and drive innovation in the industry, the two studied companies recognize the need to shift away from their previous innovation approaches characterized by long development cycles primarily focused on incremental innovation. However, despite their ambition, both companies remain predominantly focused on incremental innovation, potentially impeding the advancement of radical innovation. The transition from incremental to radical innovation can introduce conflicts within the existing system. However, this does not necessarily imply that managing both types of innovation within the same system is impossible. It is worth investigating whether certain elements within the system can be shared or should be separated, such as different operational approaches or follow-up processes. The results indicate that both types of innovation benefit from possessing capabilities related to all elements. Therefore, an alternative could be to manage both types of innovation together at a higher level, where goals, priorities, and planning align, but they may require distinct management approaches at a lower, more detailed level. In other words, the system's elements may remain the same, but the focus and direction may differ. Both types of innovation contribute to the overall function of innovation and value creation but in distinct ways. The literature emphasizes the need for both types to sustain continuous value and relevance (O'Reilly and Tushman, 2004). However, the system must create conditions that support and facilitate the management of these different types of innovation, rather than applying the same approach to both, as they are inherently different (McDermott and O'Connor, 2002; Reinertsen and Shaeffer, 2005; Chang *et al.*, 2012). It is essential to consider the impact of the entire system on both types of innovation and their respective goals. This can be challenging when including them in the same system or subsystem at a lower level, as conflicts may arise. Eisenhardt et al. (2010) emphasize the need to prioritize radical innovation when managing both radical and incremental innovation within the same system, especially for established companies that tend to lean towards incremental innovation. However, this raises another question regarding the feasibility of accommodating incremental innovation within a system designed for radical innovation.

To summarize, comprehensiveness of capabilities seems desirable for both types of innovation, although different capabilities within elements might be required. Coherence between the elements also highlights potential obstacles when system elements are interdependent. Actions taken in one element to support incremental innovation without considering how it benefits both types of innovation can have undesirable consequences for radical innovation. Correspondence suggests that the two types of innovation may need to be disaggregated or separated when it comes to goal setting. In this study, when

examining the IPOs used, it can be observed that incremental and radical innovation fall at different points on the scale within each objective. Although this might vary with different IPOs, it is likely a common phenomenon, as managing the two types requires different approaches, and they yield different outcomes. While overarching IPOs can be employed, breaking them down into different levels of IPOs or quotas (similar to a portfolio approach) may prove more effective.

6.3 Limitations

The study adopted the system elements from the international standard for innovation management systems, ISO 56002, however, capturing the system elements poses an inherent challenge due to their multitude and likely interconnections, followed by the additional difficulty of capturing them within a limited timeframe. Consequently, the study provides a limited and less detailed understanding of each element, which may not capture the full complexity of the phenomenon under investigation. Furthermore, when studying the differences between the management of incremental and radical innovation, the emphasis was primarily on providing a general description and distinction between these two types. As a result, the analysis did not extensively explore specific differences within each type, such as the various dimensions of novelty that could exist within incremental and radical innovation respectively. For instance, the distinctiveness of entering a new market compared to introducing a new offering was not explored in depth.

In terms of comprehensiveness, the study is unable to determine which elements of innovation management that are the most important. The data suggests that different factors contribute to success, and these factors that were mentioned during the interviews may vary depending on what role the informant had. Therefore, it is challenging to pinpoint the key elements in managing innovation effectively. Regarding coherence, the study does not explore the logical connections between elements or concepts. It raises the question of whether it would be valuable to examine the logical relationships that may not be evident from the mapping of the concepts in the illustrations. Correspondence was examined based on the goal of achieving more radical innovation, which may not reflect the aspirations of all organizations.

Furthermore, it is important to acknowledge that the study primarily focuses on large, established companies, which may limit the generalizability of the findings to other organizational types or contexts. The findings imply that the two companies predominantly focus on product and process development and innovation, potentially excluding other relevant areas of innovation such as service or business model innovation and how they are influenced by the dimensions of systemicity. However, it is worth noting that the choices made regarding the specific context of the study were intentional, aiming to highlight contextual aspects and facilitate analysis while pointing out the parts influenced by these aspects. For instance, the selection of innovation performance objectives (IPOs) used to analyse correspondence was based on the goals specific to the industrial context under investigation. While these objectives might be shared across multiple industries and sectors due to the widespread importance of sustainability, it is essential to recognize the contextual basis for their selection in this study. By

acknowledging the limitations and clarifying the rationale behind the choices made, the study maintains transparency and provides insights into how contextual factors influenced the research design. This helps readers understand the scope and applicability of the findings while promoting further exploration in diverse organizational settings and innovation domains.

6.4 Towards a Tentative Framework for Analysis and Evaluation of Innovation Management Systemicity

Based on the results, which indicate the relevance of comprehensiveness, coherence, and correspondence, this chapter introduces a tentative framework for analyzing system dynamics, obstacles, and opportunities in innovation management. This framework is to a large extent based on the methods employed in the analysis, inspired by Slack and Lewis (2017). Moreover, suggestions on additional aspects to consider which were not included in this study are highlighted. The framework can be used for various purposes and by both researchers and practitioners. Its specific application depends on the intended purpose and therefore, detailed comments on specific applications will be addressed where relevant.

This study adopted system elements from the international guiding standard for innovation management systems ISO 56002 (The International Organization for Standardization, 2019). However, alternative categorizations of system elements can be found in the literature. Previous studies have also highlighted differences across sectors and types of organizations (Tidd, 2021; Bessant, 2022; Kihlander and Gravenius, 2022), suggesting that different system elements may be beneficial in different contexts. Nevertheless, the results from this study as well as previous research (e.g., Hyland, Karlsson and Kristiansen, 2022) indicated that having a common language and understanding of innovation supports innovation. Given that systemic innovation involves collaborations among actors in the value chain or across different parts of the larger system (Midgley and Lindhult, 2021), it can be argued that using the same framework as a starting point could facilitate such collaborations.

To effectively analyse and evaluate an organization's innovation capability from a systems perspective, it is important to establish clear objectives for their innovation efforts. A valuable starting point could therefore be to define Innovation Performance Objectives (IPOs) that align with the overall goals of the organization. In this study, the analysis of companies' innovation capabilities was based on the overarching objective regarding the sustainability transition and the need for radical innovation. As a result, the IPOs identified were time, risk-taking, flexibility, novelty, and cost. These IPOs can be expanded or modified as necessary, depending on what the organization aim to achieve.

This study evaluated comprehensiveness by assessing the current maturity levels of innovation capabilities across different elements. Although this could provide relevant insights into the present state of organizations, it does not necessarily indicate the desired maturity levels. Therefore it could be relevant to consider the desired maturity level for comparison, similar to what is referred to as target maturity level in the self-assessment model developed by de Carvalho Botelho Santos and Ludovico de

Almeida (2022), as achieving a high maturity level in all elements may not always be the desired objective. Figure 6.1 provides an example on how this could be visualized. Note that the example does not provide a target maturity level that is desirable for all organizations.

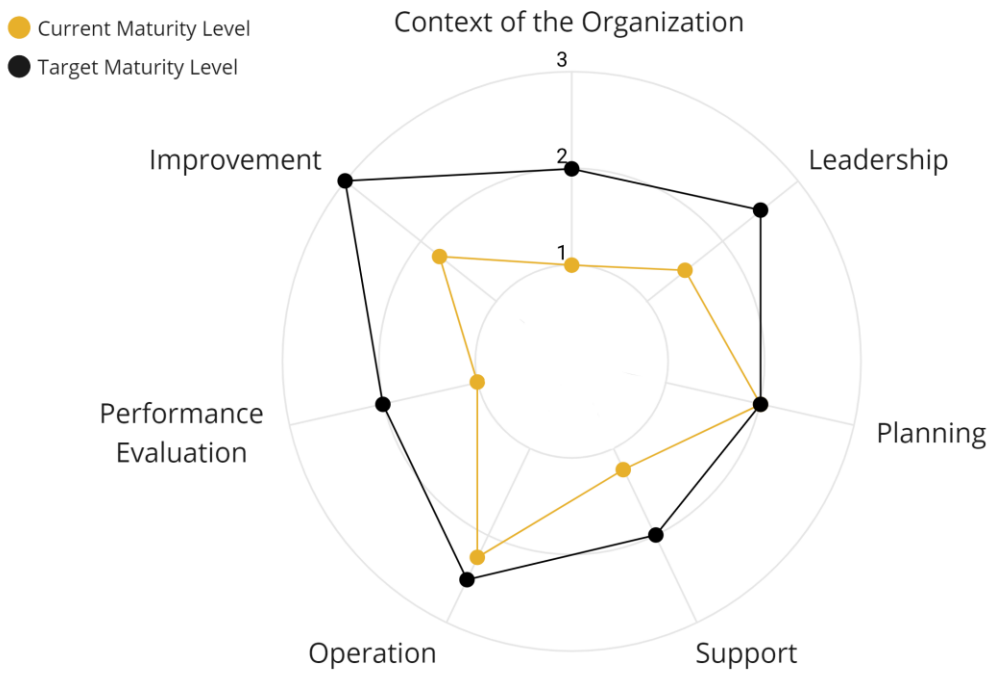


Figure 6.1. Example of target maturity level compared to current maturity level for each element. The maturity level can range between 1 and 3, where Level 1: Nothing, informal, or ad hoc, Level 2: Managed at basic level, Level 3: Defined and managed.

Furthermore, regarding comprehensiveness, it is necessary to include questions or factors to analyze and evaluate the maturity level of capabilities. The elements of the standard used in this study represent clusters of capabilities or high-level capabilities, and it may be necessary to break them down into more detailed capabilities, similar to the approach used in the study to formulate interview questions (see Appendix A). An example of this is breaking down the element Context of the Organization into more detailed capabilities such as Culture, Collaboration, Understanding the Needs of Interested Parties and Understanding the Organization. This could be illustrated in a radar chart similar to the one in Figure 6.1 to better understand the maturity levels on a more detailed level within the elements.

If using formulation of concepts, problems or strengths in the analysis, which was employed in this study and may be more common in research, these can be mapped to the different elements in accordance with Figure 6.2, where an example including five concepts (numbered 1-5) is visualized. This can provide an overview of the spread of capabilities as well as support the analysis of coherence between capabilities and/or elements.

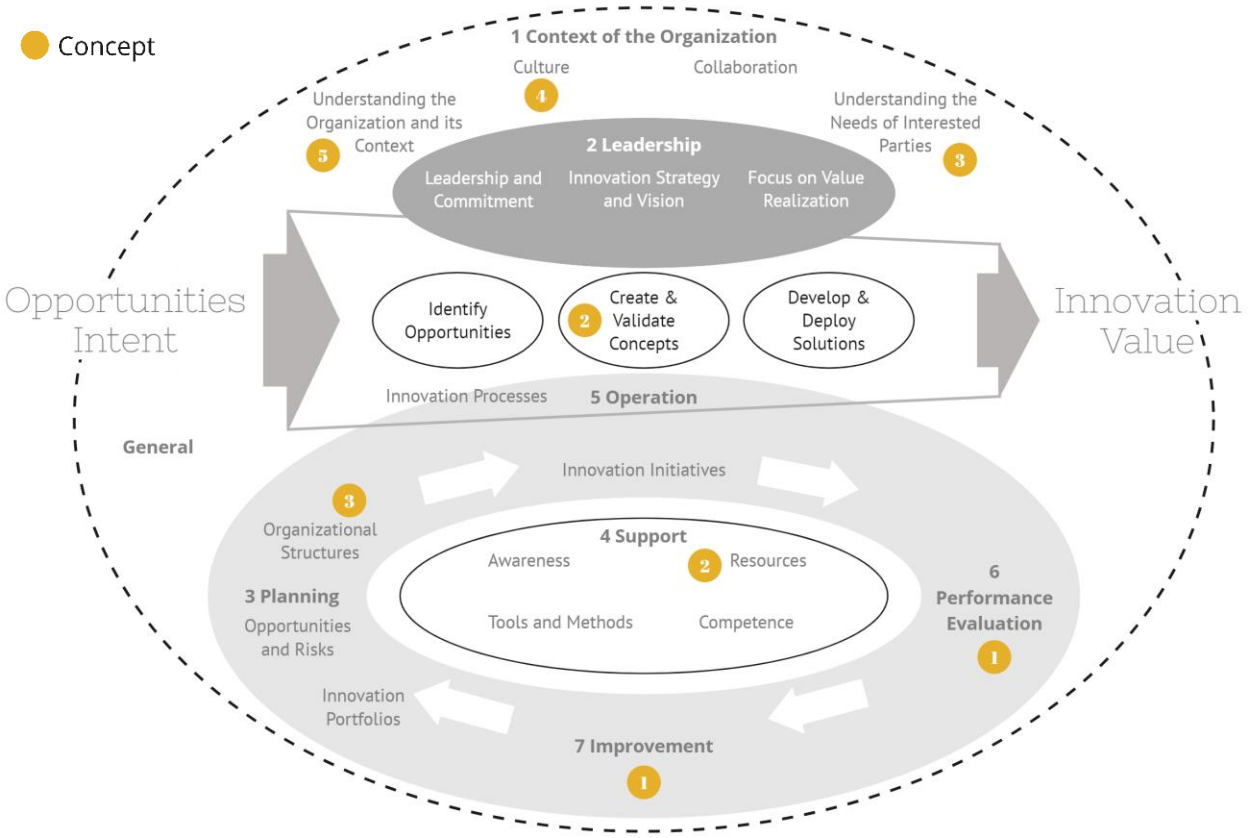


Figure 6.2. Example of a mapping of an organization's innovation capabilities.

Assessing coherence between elements can provide valuable insights into the interconnections among innovation capabilities, enabling a better understanding of their impact. This analysis can help identify how changes in one element affect other parts of the system and shed light on the underlying challenges and problems. By examining challenges in relation to the elements and how the relationships between elements affect these challenges, it becomes possible to uncover unintended consequences or broader implications that may have been overlooked. To gain a better understanding of which elements have an undesired impact on others, visualizing this relationship can be valuable. Figure 6.3 provides a visual representation of these interconnections as well as areas where additional focus could be required. Note that it is an example and does not indicate true relationships of an organization. To achieve a more nuanced and complete understanding, it is suggested to also include logical connections into the analysis that may not be explicitly evident in the mapping of concepts or problems. By considering additional logical connections, the analysis can capture a broader range of interdependencies, contributing to a more holistic perspective. This involves identifying and exploring potential relationships or linkages between different concepts, problems, or factors that may not have been initially apparent.

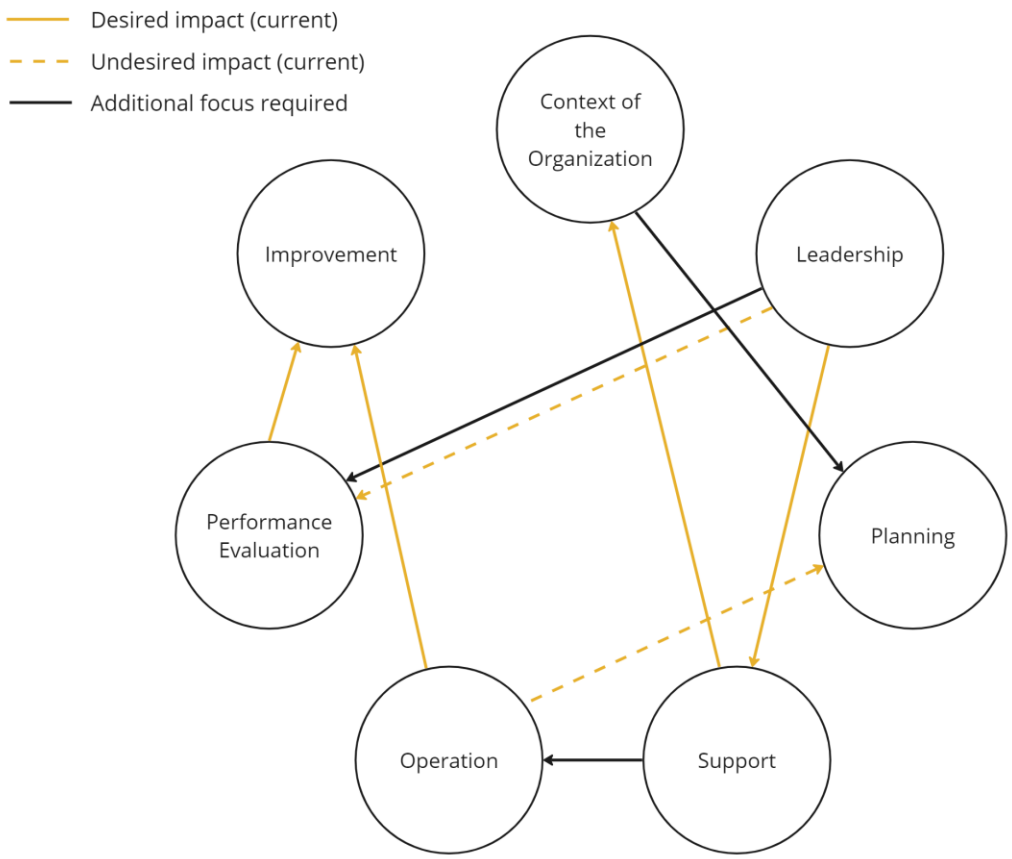
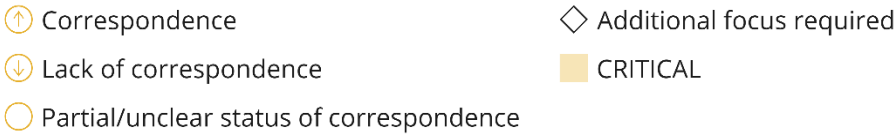


Figure 6.3. Example of coherence with current desired impact, current undesired impact as well as relationships where additional focus is required.

In terms of correspondence, it could be relevant for organizations to consider not only the current state but also where they want to improve. Mapping the areas of correspondence and identifying areas with no correspondence can provide valuable insights into where efforts need to be directed. Furthermore, including additional aspects to determine the desired state can be beneficial, an example of this is visualized in Table 6.1. Note that it does not indicate any general desired areas where additional focus is required.

Further regarding correspondence, it is possible that there are numerous intersections between the activities performed or current capabilities within elements and performance objectives. This observation was noted in the two cases examined in this study. This can pose challenges for companies in determining how to effectively utilize the results. One way to enhance the utilization of the findings, could be to establish a prioritization of objectives, allowing for the sequential identification of critical intersections that require attention when identifying areas for improvement or where focus should be directed. A similar factor, referred to as criticality, is incorporated in the Operation Strategy Matrix framework developed by Slack and Lewis (2017). Considering criticality as an additional factor in relation to correspondence could further enhance the analysis also for innovation management systems. This could be visualized in the same table as the correspondence mapping and focus areas (Table 6.1). Note that it is an example and not generalizable critical areas.

Table 6.1. Example of correspondence mapping with additional focus areas to consider and critical IPOs for each element. Explanation of symbols: arrow up = correspondence, arrow down = lack of correspondence, empty circle = partial correspondence or an unclear status of correspondence, nothing = data does not provide information about correspondence.



Innovation Performance Objectives (IPOs) →

System Elements ↓

	Time	Risk-taking	Flexibility	Novelty	Cost
Context of the Organization		↓	◇		
Leadership	↑			↑	○
Planning	◇	○			
Support			↓	↓	◇
Operation	○			↑	○
Performance Evaluation	↓	↑	◇		◇
Improvement	◇		↑		

6.4.1 Implementation of the Tentative Framework

To facilitate an implementation of the framework, a set of suggested questions to ask and actions to take have been developed, as outlined in Table 6.2. The questions can support an initial reflection on the system approach of innovation management. For questions that do not have predefined actions to take, a more reflective approach is encouraged where it could be beneficial to note down the answers or insights obtained during the reflection process. To further complement the questions and actions to take outlined in the table, filling out the spreadsheet in Appendix B could provide an overview of all steps and support the analysis of innovation capabilities. While implementing the framework may be extensive and challenging, it could be beneficial for organizations since the deeper exploration of the innovation capabilities could potentially increase the understanding of the current position and future aspirations.

Table 6.2. Questions to ask and actions to take when implementing the tentative framework.

Step	Question to ask	Action to take	Area
1	What framework should be used for the analysis?	Identify framework (e.g., ISO 56002 standard or other IMS)	Defining the System
2	What is/are the goal/-s with the innovation work?	Break down the goal/-s into IPOs	Setting IPOs
3	Does the innovation capability exist?	Map out the capability to relevant elements of the framework	Comprehensiveness: Spread
3.1	What are the consequences of absence of innovation capability?		
3.2	How do the capability support the innovation work?		
4	How well developed and managed is the capability?	Use maturity levels and illustrate this in a radar chart	Comprehensiveness: Depth
4.1	What does the current management of the capability result in?		
4.2	What is the target maturity level?	Use maturity levels and illustrate this in a radar chart	

Step	Question to ask	Action to take	Area
5	How do capabilities in the areas/elements affect capabilities in other parts?	Illustrate by drawing connections between elements (and what type of connection)	Coherence
6	What are the prerequisites for the capability to exist?		
7	In what areas/elements are capabilities missing? What are the reasons?	If needed, illustrate by drawing connections between elements where additional focus is required	Coherence
8	Are there areas/elements where capabilities exist, but are not fully utilized? What are the reasons?	If needed, illustrate by drawing connections between elements where additional focus is required	
9	Does the capability align with the IPOs or not?	Fill in type of correspondence in correspondence table	Correspondence
9.1	Is alignment desirable?		
10	Where should additional focus be placed	Illustrate in correspondence table	
11	Is correspondence status critical?	Illustrate in correspondence table	

7 CONCLUSIONS

This chapter focuses on summarizing the conclusions related to the two research questions addressed in the study. Additionally, it highlights the theoretical contributions and practical implications derived from the research findings. Furthermore, future work is also covered.

The purpose of the study was to explore and further advance the knowledge regarding innovation management from a systems perspective by studying how systemicity and its dimensions influence an organization's innovation capabilities and investigating these dimensions in relation to incremental and radical innovation respectively. Through concluding the findings of the study in the following sections, thereby achieving the purpose.

> RQ 1: How does innovation management systemicity influence an organization's innovation capabilities, where systemicity is defined by the three dimensions (1) comprehensiveness, (2) coherence, and (3) correspondence?

Related to research question 1, the study suggests that having capabilities related to all system elements is relevant to the overall innovation capability of organizations. However, the suitable maturity level for each element or its components remains uncertain. Furthermore, the findings highlight that all elements are interconnected, demonstrating their interdependence within the system. The companies demonstrated capabilities in various elements, but these capabilities were dependent on other elements, underscoring the significance of coherence for innovation capability. The study suggests that establishing correspondence between capabilities, innovation efforts, and overarching goals is essential for enhancing overall innovation capability. Alignment between capabilities and goals facilitates the achievement of the goals. While the findings suggest that organizations can benefit from exhibiting systemicity through its dimensions comprehensiveness, coherence, and correspondence in their innovation management approaches, the extent to which high levels of systemicity are desirable remains uncertain. It is unclear whether striving for maximum systemicity in innovation management approaches is always advantageous, as innovation inherently involves novelty, variations, and constant change. Consequently, management approaches should accommodate and facilitate these dynamic aspects, and excessively high levels of systemicity may impede the variation and adaptability necessary for innovation. Further research is needed to determine the suitable balance between systemicity and flexibility in order to effectively support and enhance innovation capabilities within organizations.

> RQ 2: What does systemicity indicate regarding hindrances and opportunities for managing incremental and radical innovation within the same system?

Related to research question 2, the study indicates that systemicity plays a significant role in understanding the hindrances and opportunities associated with managing incremental and radical innovation within the same system. Comprehensiveness underscores the importance of distinct management approaches and considerations, even though the essential elements for both incremental and radical innovation may not differ significantly. This suggests the need for capabilities in all areas

while recognizing the different handling required for each innovation type. The system's components may remain the same, but the focus and direction may differ. Coherence emphasizes the potential obstacles resulting from interdependencies between system elements being overlooked, highlighting the importance of considering the impact on both types of innovation. Correspondence further suggests that goal setting may require separate consideration for incremental and radical innovation, as they have distinct objectives and outcomes. Managing the two types of innovation necessitates different approaches and recognition of their unique positions within each objective.

7.1 Theoretical Contributions

The study's theoretical contributions stem from the application of theories from other domains, namely the definition of systems from systems theory and the Operations Strategy Matrix developed by Slack and Lewis (2017), to the field of innovation management systems. By investigating the relevance and usefulness of these theories in the context of innovation management, the study expands the theoretical understanding of how established frameworks can be applied and adapted to address the specific challenges and dynamics of innovation. This mix of theories not only enriches the field of innovation management but also highlights the interconnectedness and transferability of knowledge across different domains, fostering interdisciplinary perspectives and opening new avenues for theoretical development.

The study also provides a novel approach to studying the systemic aspects of innovation management approaches and the study's insights can guide future research in the field, encouraging further exploration of the systemic aspects of innovation management and their impact on organizational performance. By introducing the concept of systemicity and its dimensions in the context of innovation management, the study offers a framework for analyzing and understanding how different elements of an innovation management system interact and align with the overall purpose. These dimensions shed light on the extent to which the system exhibits systemic characteristics and functions as a cohesive whole. Furthermore, the study also identify and discuss potential challenges as well as opportunities and hinders with systemic approaches to innovation management. Lastly, the study does not only introduce systemicity as a new concept for capturing how systemic a system is, but also provides a way of measuring systemicity which can support future quantitative studies within the field of innovation management.

7.2 Practical Implications

The study's findings have practical implications for managers and those involved in developing innovation support frameworks within organizations. Firstly, it emphasizes the importance of considering the systemic nature of innovation management. Managers should pay attention to comprehensiveness, coherence and correspondence in their approach by ensuring that all relevant elements are adequately addressed. This includes assessing the spread and depth of capabilities within the system, analysing their interdependencies, how this impact the overall capability and how the

capabilities contribute to the organizational objectives for innovation. By adopting a systems perspective, managers can better identify potential obstacles and opportunities, allocate resources effectively, and make informed decisions to enhance overall innovation performance.

The research outcomes also have implications for standard development organizations, researchers, and companies utilizing innovation management standards. The study's findings shed light on the significance of incorporating systemicity into existing standards or frameworks. Feedback can be provided to standard development organizations, highlighting the need to consider the three dimensions of systemicity: comprehensiveness, coherence, and correspondence. This feedback can help refine and improve existing standards, ensuring that they align with the complexities of managing innovation within a system. It is important to consider the degree of systemicity that is necessary, recognizing its significance while avoiding excessive rigidity that could potentially impede the emergence of new aspects appearing in the system. The International Organization for Standardization (ISO) is currently developing a certifiable innovation management system standard (The International Organization for Standardization, n.d.). It is important to carefully assess the trade-off between the advantages of standardization and the requirement for flexibility in innovation management practices. Standardization should support, not hinder, organizations in their innovation efforts. A certifiable standard need to allow also for the system to change and adapt, as this is fundamental to innovation.

7.3 Future Work

Expanding upon the discussion and prior conclusions, there are specific areas that that would benefit from improvement or further investigation to advance the understanding and application of systemic approaches to innovation management.

One important area for future research is to explore different types of organizations including those that explicitly adopt a systemic approach or utilize standards and frameworks to analyze systemicity in their contexts. Investigating these companies can provide deeper insights into the effectiveness of systemicity and shed light on whether a high degree of systemicity is desirable. Additionally, examining organizations from various sectors and industries would contribute to the understanding of how systemicity manifests in different contexts and regarding different forms of innovation (e.g., product, service, or process innovation).

Another avenue for future work is to investigate established companies that have consistently demonstrated proficiency in both incremental and radical innovation. Understanding how these companies manage and balance both types of innovation can offer further insights into opportunities, challenges and hence, considerations to take in relation to systemic innovation approaches to enhance innovation performance.

As a final remark, further research could also delve into the examination of how the identified innovation capabilities and their systemicity influence organizations' innovation performance, as it has

not been covered in this study. This exploration would likely require a more extensive and in-depth study, involving the application of additional research methods.

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APPENDIX A: INTERVIEW GUIDE

This appendix presents the interview guide used in the study. Words and sentences in italic are to support for what to listen for or examples of follow-up questions.

Introduction

Information about the interview

Ask about recording.

We will start with questions about you, your role and how it is related to the innovation work at the company, then move over to questions about the innovation intent and a summary of how innovation is managed. Most of the time will be spent on questions about different parts of how you manage innovation, such as innovation strategy, processes, planning, and evaluation. With innovation work we consider all parts contributing to innovation, from intent to outcome/results.

About the informant

What is your **role** within the company and what is included in your **tasks and responsibilities**?

- How is that related to the innovation work?
- How often are you in contact with the innovation work?
- What parts are you working with (strategy, processes etc)

Innovation Intent

(Q1) How would you with two sentences **define the term innovation**?

- It that how the term is used at the company?
- Are you using other terms?

(Q2) What are the **driving forces** behind your innovation work?

- What needs of innovation do you have?
  Technology/market/customer-driven?
- What do you want to achieve with your innovation work?
  What value does it create?

(Q3) Do you distinguish between **different types** in terms of **degree** of innovation? (major/small changes)

- What terms do you use?
- *If yes:* How do you manage these types?
  - How are the resources distributed (approximately)?
  - Are they handled in the same working groups, the same ways of working? Or separately?

    - How well does it work? Outcomes for different types of innovation?
- *If no:* Why not? How do you do then?

How is the company performing?

(Q4) Can you briefly describe how you **manage your innovation work**?

(Q5) Does your innovation work **continuously result in value**?

(Q6) How would you say you **perform compared to competitors** when it comes to innovation?

System elements

Leadership

(Q7) How are **roles and responsibilities** around the innovation work structured?

(Q8) Is there a **stated vision** and **concrete goals** for the innovation work? *Is there a difference in goals and vision regarding different degrees of innovation?*

- How is it **communicated** internally and externally?

(Q9) Is the **top management involved** in the innovation work? How?

- How do they **show commitment**? How do they act? Examples?

Context (& Support)

(Q10) How do you work to understand **external trends and drivers**?

- Is it done continuously?
- How is that knowledge used?

(Q11) Can you describe how you work with external **collaborations**?

- How do you make use of and share knowledge outside the organization?

(Q12) How do you work to understand and make use of your **internal abilities and resources**?

- How do you make use of and share knowledge within the organization?
- (S) How do you ensure that you have the necessary skills and resources available to enable your innovation work?

Operations/Processes (& Context)

(Q13) Can you describe how you work with **planning, implementation and follow-up of innovation initiatives** such as innovation projects, programs or other types of coordinated activities?
Listen to how they work with the different "parts" of the process. How do they handle ideas? Identify opportunities? Develop concepts? Test concepts? Develop solution? Introduce solutions?

- Can you **adapt processes and structures** according to the type of innovation if you see that it is needed?
    - If so, **how** is that adjustment accomplished?

(Q14) How do you work with **creativity** in your innovation work?

- (C) Would you say that creativity can coexist with **efficiency** in your company? How? Examples?
- (C) How is this relation reflected in the **work culture**? (Dare to take risks etc.) *Does the work culture create room for creativity and efficiency at the same time?*

Planning (& Support)

(Q15) How do you implement/**realize** your **goals** in practice? Selection of projects, etc *Differences incremental/radical?*

- How are the **goals anchored** in the organization?
- Are **funds/resources allocated** to meet the goals?
- (S) Are **resources distributed** in such a way that there is a clear connection to innovation goals and vision?

(Q16) What **uncertainties and risks** are you trying to manage today in your innovation work? Examples?
If innovation is about renewal where uncertainty and risk are a given part, what do you think about that?

- How do you **manage** these uncertainties and risks?

(Q17) Do you work with **innovation portfolios**? How?

- How does it affect and contribute to other parts of your innovation work?

Performance Evaluation & Improvements

(Q18) How do you **follow up and evaluate** your innovation work, i.e., the goals you set up?

- (PE) How do you know you are moving towards your goals?
- (PE) How do you **measure** it? Do you use any metrics?

(Q19) How do you work to **improve** your innovation work?

- (I) How do you make improvements **based on the evaluation** of the results?
- (I) How do you **continuously improve** your innovation work?

Systems approach

(Q20) If we look at all the parts needed for innovation work, in which areas are you **strong and weak**? (from intent to output)

- How does it play out?
- If you have problems/weaknesses, **why** do you have those problems?

(Q21) If we look at all the parts needed for innovation work again, which **parts** are most **important** for you to succeed?

How is the company performing? (again)

Would you say that the innovation work results in the **intended value**?

Finishing questions

Is there something more that's connected to the innovation work that you want to add?

Who do you think might be relevant for us to interview, with knowledge in all or parts of what we have talked about?

May we contact you if there is anything else we are wondering about or if there is something we would like clarified? In that case, do you prefer to be contacted by phone or email?

APPENDIX B: TOOL TO SUPPORT ANALYSIS AND EVALUATION OF IMS

This appendix presents a tool to support the implementation of the tentative framework for analysis and evaluation of innovation management systems. The notes in the spreadsheet serve as an illustrative example. Please note that the spreadsheet provided is the initial version and may undergo further updates and refinements in the future.

Defining the System		Comprehensiveness						Coherence																														Correspondence				
			Maturity Level [X] Level 1: Non-existent, informal or ad-hoc Level 2: Managed at basic level Level 3: Defined and managed						Desired impact [X], Undesired impact [O], Additional focus required □ The row affect the column.																														↑ Correspondence ↓ Lack of correspondence ◯ Partial or unclear status of correspondence □ Critical				
								Element																																		
Element	Innovation Capability	Existing Capability [Y/N]	Current Maturity Level			Target Maturity Level			Context of the Organization				Leadership				Planning				Support				Operation			…	Performance Evaluation			…	Improvement			…	Innovation Performance Objective (IPO)				
			1	2	3	1	2	3	C1	C2	C3	C4	L1	L2	L3	L4	P1	P2	P3	P4	S1	S2	S3	S4	O1	O2	O3		PE1	PE2	PE3		I1	I2	I3		Time	Risk-taking	Flexibility	Novelty	Cost
Context of the Organization	(C1) Understanding the Needs of Interested Parties	Y		X				X					X						O																	↑				
	(C2) Understanding the Organization and its Context	Y		X				X										X																				↓	↑	
	(C3) Collaboration	Y	X				X							X													O										◯				
	(C4) Culture	N	X					X																																		
Leadership	(L1) Innovation Strategy and Vision	Y		X			X													X														X				↑				↓
	(L2) Opportunities and Risks	Y			X			X																			X													◯	
	(L3) Innovation Portfolios																																								
	(L4) Organizational Structures																																								
Planning	(P1) Competence																																								
	(P2) Resources																																								
	(P3) Tools and Methods																																								
	(P4) Awareness																																								
Support	(S1) Competence																																								
	(S2) Resources																																								
	(S3) Tools and Methods																																								
	(S4) Awareness																																								
Operation	(O1) Operational Planning and Control																																								
	(O2) Innovation Initiatives																																								
	(O3) Innovation Processes																																								
	…																																								
Performance Evaluation	(PE1) Monitoring, Measurement, Analysis, and Evaluation																																								
	(PE2) Internal Audit																																								
	(PE3) Management Review																																								
	…																																								
Improvement	(I1) General																																								
	(I2) Deviation, Nonconformity, and Corrective Action																																								
	(I3) Continual Improvement																																								
	…																																								

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